J ²

FZl_fZlbq_kdb (63 iZd_l ij_^klZ\ey_lky ke_^mxsbf h[jZahf

(63 3DFNHW (63 +HDGHU G (63 3D\ORDG G (63 7UDLOHU G (63

l j b f _ q Z g >bZe_² kqblZ_lky qlh[u ihemqZl_ey (63 iZd_lZ « (63 ,&9 y\eyxlky [Zclh\ufb ij_^klZ gbyfb agZq_gbc]jmZ_ESPHeader «ESPICV khhl_lkl_ggh 63 iZd_l 61 « ,&9 y\eyxlky [Zclh\u fb ij_^klZ\e_gbyfb agZq_gbc ihe_ICVSPkhhl_lkl_ggh

B^_glbnbdZlhj aZsbs_ggh]h kh_^bg_gby 63,

B^_glbnbdZlhj aZsbs_ggh]h kh_^bg_gby 63, kem`bl mgbdZevguf b^_glbnbdZ g_ ihemqZl_ey (63 iZd_lZ lhe_ kh^_j`bl ijhba\hevgh_ [blgh_ agZq_gb_ AgZ g_ ² ijb ij_^klZ\e_gbb wlh]h ihey \ \b^_ [blgh]h qbkeZ \ k_l_\hf ihjy^d_ jh\Zgu ^ey ki_pbZevguo p_e_c b g_ ^he`gu bkihevah\Zlvky \ ,3VHF Djhf_ lh]h aZij_s_gguf ^ey 63, b g_ ^he`gh bkihevah\Zlvky

AgZq_gb_ 63, \u[bjZ_lky gZ wlZi_ kha^Zgby 6$ lZdbf h[jZahf qlh[u ihemqZl h^ghagZqgh hij_^_eblv d dZdhfm aZsbs_gghfm kh_^bg_gbx hlghkblky ihemq_g _lky qlh[u agZq_gby 63, \u[bjZebkv g_ij_^m]Z^u\Z_fh ^ey \g_rg_]h gZ[ex^Zl iZju (63 6$ ijhlhdhehf ,.(Y dZ`^uc ba mqZklgbdh\ \u[bjZ_l 63, ^ey lh]h aZsbs gby \ dhlhjhf hg [m^_l y\eylvky ihemqZl_e_f iZd_lh\ b ^_eZ_l wlh lZdbf h[jZa q_gbb (63 iZd_lZ hg \k_]^Z fh] h^ghagZqgh gZclb (63 6$ ih 63,

lhjy^dh\uc ghf_j iZd_lZ 61

lhjy^dh\uc ghf_j iZd_lZ 6HTXHQFH 1XPEHU ij_^klZ\ey_l kh[hc [_aagZdh\ \ k_l_\hf ihjy^d_ [Zcl l_j\uc iZd_l ihkeZgguc ih \gh\v kha^Zgghfm aZsbs_gghf ^he`_g bf_lv ghf_j dZ`^uc ihke_^mxsbc iZd_l ² gZ _^bgbpm [hevr_ lhjy^ d_lZ g_ ^he`_g i_j_ihegylvky lh _klv _keb g_ kh]eZkh\Zg jZkrbj_gguc ihjy^dh hlkuedb 32 ± iZd_lh\ ^Zggh_ aZsbs_ggh_ kh_^bg_gb_ g_ ^he`gh ^Ze__ bkiheva

JZkrbj_gguc ihjy^dh\uc ghf_j iZd_lZ (61

ljb g_h[oh^bfhklb i_j_^Zqb \ jZfdZo h^gh]h aZsbs_ggh]h kh_^bg_gby [he__ fh`_l [ulv bkihevah\Zg jZkrbj_gguc ihjy^dh\uc ghf_j iZd_lZ ([WHQGHG 6HTX ihevah\Zgb_ jZkrbj_ggh]h ghf_jZ kh]eZkh\u\Z_lky Z[hg_glZfb gZ wlZi_ kha^Zgb ihjy^dh\uc ghf_j ij_^klZ\ey_l kh[hc [blgh_ [_aagZdh\h_ qbkeh Lhevdh feZ^ ^Zxlky \ iZd_l_ \ ihe_ 6HTXHQFH 1XPEHU klZjrb_ [blZ ojZgylky Z[hg_glZf 6$ ijb wlhf hgb \dexqZxlky \ \uqbke_gb_ bfblh\klZ\db ljb bkihevah\Zgbb jZk dh\h]h ghf_jZ ihke_ hlkuedbiZd_lh\ ^Zggh_ aZsbs_ggh_ kh_^bg_gb_ g_ ^he`gh ihevah\Zlvky

H[jZ[hldZ ihjy^dh\h]h ghf_jZ iZd_lZ gZ ijb_fghc klhjhg_

AZsblZ hl ZlZd ih\lhjghc i_j_^Zqb iZd_lh\ y\ey_lky hipbhgZevghc b \ g_dhlhjuc gbfhc gZijbf_j _keb bkihevam_fu_ \ 6$ ljZgknhjfu g_ h[_ki_qb\ZxHkhZckh\Z bfblh wlZ hipby g_ kh]eZkh\u\Z_lky b ihkueZxsZy klhjhgZ ijb hlkued_ dZ`^h]h iZd_l ebqb\Zlv _]h ihjy^dh\uc ghf_j ljb_fgZy klhjhgZ hij_^_ey_l ^he`_g eb h[jZ[Zlu \uc ghf_j iZd_lZ \ dhgl_dkl_ ^Zggh]h aZsbs_ggh]h kh_^bg_gby beb ^he`_g b]gl

>ey aZsblu hl ZlZd ih\lhjghc i_j_^Zqb iZd_lh\ ihemqZl_ev ^he`_g ijh\_jylv f_j dZ`^h]h \oh^ys_]h iZd_lZ b \ kemqZ_ _keb lZdhc iZd_l m`_ [ue ihemq_g hl[jZku\Zlv \gh\v ijbr_^rbc iZd_l WlZ ijh\_jdZ ^he`gZ \uihegylvky gZ kZfhf jZ jZ[hldb \oh^ysbo iZd_lh\

< ijhp_kk_ i_j_^Zqb ih k_lb iZd_lu fh]ml i_j_f_rb\Zlvky bkq_aZlv b fgh`blv ghc h[jZ[hldb ihjy^dh\h]h ghf_jZ iZd_lZ ijb_fgZy klhjhgZ ^he`gZ ih^^_j`b\Zlv ©k kh^_jZs__ bgnhjfZpbx h ihjy^dh\uo ghf_jZo ihke_^gbo mki_rgh h[jZ[hlZgguo fZevguc jZaf_j hdgZ ² ljZ\Zy ]jZgbpZ hdgZ hljZ`Z_l kZfh_ [hevrh_ agZq_gb_ f_jZ iZd_lZ ba ihemq_gguo b mki_rgh h[jZ[hlZgguo (63 iZd_lh\ E_\Zy]jZgbpZ jZdbf h[jZahf ijZ\Zy]jZgbpZ hdgZ fbgmk jZaf_j hdgZ iexk h^bg ljb kha^Zgbb abjm_lky lZdbf h[jZahf qlh e_\Zy]jZgbpZ mklZgZ\eb\Z_lky jZ\ghc _^bgbp_ ijZ Z hdgh aZihegy_lky we_f_glZfb mdZau\Zxsbfb qlh khhl_lkl\mxsbc iZd_l _s_ g k ghf_jZfb f_gvrbfb q_f agZq_gb_ e_\hc]jZgbpu hdgZ hl[jZku\Zxlky lZd_lu k xsbfb \ ]jZgbpu hdgZ hl[jZku\Zxlky _keb \ khhl_lkl\mxs_f we_f_gl_ hdgZ klhb dhc iZd_l m`_ [ue h[jZ[hlZg \ ijhlb\ghf kemqZ_ ² h[jZ[Zlu\Zxlky lhke_ mki_rgh]h iZd_lZ khhl_lkl\mxsbc we_f_gl hdgZ ihf_qZ_lky dZd h[jZ[hlZgguc lZd_lu k g

ФЕДЕРАЛЬНОЕ АГЕНТСТВО ПО ТЕХНИЧЕСКОМУ РЕГУЛИРОВАНИЮ И МЕТРОЛОГИИ

J ²

q_f agZq_gb_ ijZ\hc]jZgbpu hdgZ h[jZ[Zlu\Zxlky < kemqZ_ _keb hgb h[jZ
]jZgbpZ hdgZ mklZgZ\eb\Z_lky jZ\ghc ghf_jm ihemq_ggh]h iZd_lZ khhl_lk
]jZgbpZ

l j b f _ q Z g b y baf_g_gby \ kdhevays_f hdg_ ^he`gu \uihegylvky lhevdh ihke_ m
iZd_lZ h[yaZl_evgh \dexqZxs_c \ k_[y ijh_jdm _]h bfblh\klZ\db

Hkh[_gghklb h[jZ[hldb jZkrbj_ggh]h ihjy^dh\h]h ghf_jZ iZd_lZ gZ
Ijb bkihevah\Zgbb jZkrbj_ggh]h ihjy^dh\h]h ghf_jZ iZd_lZ ijh_jdZ gZ i
s_kl\ey_lky ZgZeh]bqgh lhfm dZd wlh hibkZgh \ aZ bkdexq_gb_f lh]
[ulv [blgufb lhkdhevdm \ iZd_l ^he`lky lhevdh feZ^rb_ [blZ jZkrbj
ghf_jZ lh ^ey ihemq_gby ihegh]h [blgh]h ghf_jZ \oh^ys_]h iZd_lZ klZjrb
wfibjbq_kdb ke_^mxsbf h[jZahf H[hagZqbf

$W_{V]}$ ² jZaf_j kdhevays_]h hdgZ \ \b^_ [bfgh]h qbkeZ 2[31]
R ² ijZ\Zy]jZgbpZ hdgZ [blZ
R_O ² feZ^rb_ [blZ R lh $R_O = k\&\ 5^{32} \pm$
R_h ² klZjrb_ [blZ R lh $R_h = k\frac{R_O}{2^{32}}\ \&(2^{32}\ 1)$
L ² e_\Zy]jZgbpZ hdgZ [blZ
L_O ² feZ^rb_ [blZ L lh $L_O = k\&\ 2^{32} \pm 1$
L_h ² klZjrb_ [blZ L lh $L_h = k\frac{L_O}{2^{32}}\ \&(2^{32}\ 1)$
S ² jZkrbj_gguc ghf_j \oh^ys_]h iZd_lZ [blZ
S_O ² feZ^rb_ [blZ S ba ihey Sequence Number \oh^ys_]h iZd_lZ
S_h ² klZjrb_ [blZ S \hkklZgZ\eb\Zxlky gZ ijb_fghc klhjhg_
<hafh`gu ^\_ kblmZpbb jZkiheh`_gby kdhevays_]h hdgZ
$R_O \bullet W_{V]} \pm 1$
$R_O \bullet W_{V]} \pm 1$
>ey kblmZpbb $S_O \bullet L_O$ k[e^b L_O 5± ; $V_{]}$ lh $S_h = R_h$ bgZq $S_h = R_h$
>ey kblmZpbb $S_O \bullet L_O$ k[e^b L_O 5± ; $V_{]}$ lh $S_h = R_h \pm$ bgZq $S_h = R_h$
>Zgguc Ze]hjblf iha\hey_l \hkklZgh\blv jZkrbj_gguc ghf_djhf_ZlZ\dgZ\bbab
iZd_lZ b ihemq_ggh]h ke_ q_]h ^he`_g bkihevah\Zlvky Ze]hjblf hibkZgguc \
gZ lh qlh \k_ _ebqbgu y\eyxlky [blgufb ljbeh`_gb_ < kh^_j`bl nhjfZel
Ze]hjblfZ h[jZ[hldb jZkrbj_ggh]h ghf_jZ iZd_lZ gZ ijb_fghc klhjhg_

l j b f _ q Z g b y
Ex[u_ baf_g_gby \ kdhevays_f hdg_ ^he`gu \uihegylvky lhevdh ihke_ mki_rgh
klZgh\e_gguf agZq_gby \ kdhevays_f hdg_ ^he`gu \uihegylvky lhevdh ihke_ mki_rgh
HibkZgguc Ze]hjblf \hkklZgh\e_gby klZjrbo [blZ jZkrbj_ggh]h ghf_jZ iZd_lZ
hkh[_gghklvx ex[hc \oh^ysbc iZd_l g_ ihiZ^Zxsbc \ hdgh kqblZ_lky gh\uf Z g_ h
keh\Zfb iZd_l hlijZ\e_gguc k jZkrbj_gguf ghf_jZ iZd_lZ jZkrbcky \ dZgZe_ gZklhevdh qlh \
ihemq_gby gZ ijb_fghc klhjhg_ kijZ_^e b ihemq_gby hibkZggh]h Ze
[m^_l \hkijbgyl gZ ijb_fghc klhjhg_ dZd iZd_l k jZkrbj_gguf ghf_jZ iZd_lZ lZdhc iZd
[m^_l hl[jhr_g g_ gZ wlZi_ ijh_jdb gZ ih\lhjguc iZd_l Z gZ wlZi_ ijh_jdb bfblh\klZ
HibkZgguc Ze]hjblf \hkklZgh\e_gby klZjrbo [blZ jZkrbj_ggh]h ghf_jZ iZd_lZ
kblmZpbb dh]^Z ijhbkoh^bl ihiejy ihke_^h\Zl_evguo iZd_lh\ Ohly lZdh_ kh[ulb_
_jhylguf _keb hgh \k_ `_ ijhbahc^_l lh \k_ ihke_^mxsb_ \oh^ysb_ iZd_lu ^Zggh]h
[m^ml hl[jZku\Zlvky ba aZ g__jgh]h S_h \hkklZgh\e_gby klZjrbo gZjmr_gby bfblhaZsblu
_fgZy klhjhgZ h[gZjm`bl qlh \ l_q_gb_ hij_^_e_ggh]h \j_f_gb \k_ \oh^ysb_ iZd_lu
ih ijbqbg_ gZjmr_gby bfblhaZsblu lh hgZ fh`_l ihiulZlvky ijh_klb j_kbgojhgbaZpb
[_j_lky ijhba\hevguc hl[jhr_gguc \oh^ghc iZd_l ^ey g_S_h \hkklZgh\e_gby klZjrbo lh
b ^Ze__ bgdj_f_glbjm_lky \ pbde_ k ihiuldhc dZ`^uc jZa ijh_jblv bfblh\klZ\dm iZd
agZq_gby S_h Dhebq_kl\h bl_jZpbc pbdeZ ^he`gh [ulv h]jZgbq_gh lhke_ bkq_jiZgby bl
kl\bb iheh`bl_evgh]h j_amevlZlZ fh`gh ih\lhjblv \kx ijhp_^mjm bkihevamy ^jm]hc h
dZdhc lh fhf_gl bfblh\klZ\dZ iZd_lZ mki_rgh ijh_jblky lh iZjZf_lju hdgZ g_h[oh^b
ihemq_ggh_ agZq_gby lh]h _keb \ ijhp_kk_ \uiheg_gby j_kbgojhgbaZpbb ijb^_l iZ
mki_rgh h[jZ[hlZg ijhp_kk j_kbgojhgbaZpbb ke_^m_l ij_djZlblv

Ijb bkihevah\Zgbb jZkrbj_ggh]h ghf_jZ iZd_lZ j_dhf_g^m_lky bkihevah
f_g__ >ey \ukhdhkdhjhklguo k_l_c beb k_l_c k ih\ur_ggufb ihl_jyfb j_d
\Zlv [dvrb_ agZq_gby

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J ²

>hihegbl_evgh_ aZiheg_gb_ 7)& 3DGGLQJ

< ijhlhdhe_ (63 fh`_l bkihevah\Zlvky ^hihegbl_evgh_ aZiheg_gb_ gZau\Z_f
7UDI¿F)ORZ &RQ¿GHQWLDOLW\ 3DGGLQJ Payload Data Padding
khdjulby bklbgghc ^ebgu i_j_^Z\Z_fuo ^Zgguo ^ey mkeh`g_gby aZ^Zqb ZgZeb
Bkihevah\Zgb_ TFC Padding y\ey_lky hipbhgZevguf b kh]eZkm_lky gZ wlZi_ kha^Zgby (
g_ ^he`gZ bkihevah\Zlvky _keb hgZ g_ [ueZ kh]eZkh\ZgZ

L_ogbq_lky TFC Padding hkms_kl\ey_lky iml_f m_ebq_gby Payload Data aZ
kq_l ^h[Z\e_gby d g_fm gZ i_j_^Zxs_c klhjhg_ ^h rbnjh\Zgby ijhba\hevguo ^Zgg
qlh[u ih j_amevlbjmxs_c ^ebg_ iZd_lZ g_\hafh`gh [ueh hij_^_eblv ^ebgm i_j_
dZd ijZ\beh \k_ iZd_lu nhjfbjmxlky h^bgZdh\hc ^ebgu GZ ijb_fghc klhjhg_ i
gby iZd_lZ wlb ^Zggu_ hl[jZku\Zxlky Kh^_j`bfh_ ^h[Z\ey_fuo ^Zgguo g_ j_]eZf

lhkdhevdm TFC Padding \ (63 iZd_l_ g_ i_j_^Z_lky lh ijbgbfZxsZy klhjhgZ ^h
hij_^_eblv]^_ aZdZgqb\Zxlky bkoh^gu_ ^Zggu_ b gZqbgZ_lky aZiheg_gb_ dh
Wlh fh`gh k^_eZlv lhevdh _keb \ kZfbo bkoh^guo ^Zgguo \ y\ghf \b^_ ijbkmlkl\
ghklb TFC Padding fh`gh bkihevah\Zlv

\ lmgg_evghf j_`bf_

\ ljZgkihjlghf j_`bf_ ijb mkeh\bb qlh ljZgkihjlguf ijhlhdhehf y\ey_lky 8'3
beb ^jm]hc ijhlhdhe y\gh kh^_j`Zsbc ^ebgm i_j_^Z\Z_fuo ^Zgguo

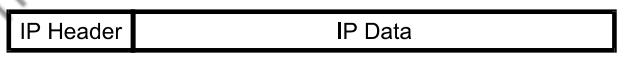
Bkihevah\Zgb_ aZiheg_gby TFC Padding y\ey_lky hipbhgZevguf \ ijhlhdhe_ (63 >jm]bf
pbhgZevguf kihkh[hf ^hklb`_gby k\hckl\Z 7)& y\ey_lky i_jbh^bq_kdZy hlkuedZ
iZd_lh\ g_ g_kmsbo ihe_aghc bgnhjfZpbb gh k lhqdb aj_gby klhjhgg_]h gZ
fuo hl ^jm]bo (63 iZd_lh\ LZdb_ iZd_lu ^he`gu bf_lv dhjj_dlgu_ agZq_gby \k_
Payload Data dhlhjh_ fh`_l kh^_j`Zlv ijhba\hevgu_ ^Zgguo Next Header lhe ey lZdbo iZd_lh\
^he`gh bf_lv agZq_gb_ GZ ijb_fghc klhjhg_ (63 iZd_lu dhlhju_ ihke_ jZkrb
`Zl agZq_gb_ \ ihe_ 1H[W +HDGHU ^he`gu hl[jZku\Zlvky [_a ^Zevg_cr_c h[j

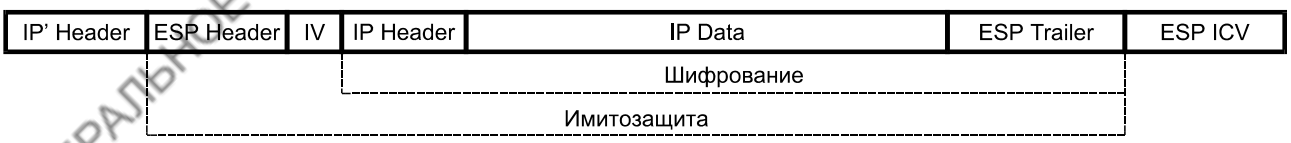
J_`bfu ijhlhdheZ (63

ljhlhdhe (63 fh`_l [ulv bkihevah\Zg \ ^\mo j_`bfZo ² lmgg_evghf b ljZgkihjl
guc j_`bf y\ey_lky iZjZf_ljhf (63 6$

Lmgg_evguc j_`bf

< lmgg_evghf j_`bf_ (63 aZsbsZ_l _kv bkoh^guc iZd_l \dexqZy _]h aZ]he
oh^guc aZsbsZ_fuc ,3 iZd_l p_ebdhf ihf_ylZ_lky Payload Data \ (63 iZd_lZ \f_kl_ kh
k\hbf ,3 aZ]heh\dhf kf jbkmghd lhke_ wlh]h nhjfbjm_lky gh\uc ,3 iZd_l k g
\dexqZxsbf gh\u_ agZq_gby ,3 Z^j_kh\ h[uqgh ^jm]b_ q_f \ bkoh^ghf ,3 aZ]h
]jmadhc gh\h]h ,3 iZd_lZ klZgh\blky (63 iZd_l < lmgg_evghf\j_`bf_ Next Header
^he`gh bf_lv agZq_gb_ ^ey ijhlhdheZ ,3Y beb agZq_gb_ ^ey ijhlhdheZ ,3
e_aghc gZ]jmadhc (63 iZd_lZ y\ey_lky ,3 iZd_l GZ ijb_fghc klhjhg_ ihke_ h[
\g_rgbc ,3 aZ]heh\hd hl[jZku\Z_lky





Jbkmghd ² Lmgg_evguc j_`bf bkoh^guc b aZsbs_gguc iZd_lu

< lmgg_evghf j_`bf_ ^himklbfh ijbf_g_gb_ (63 d njZ]f_glbjh\Zgguf ,3 iZd_lZ
^himkdZ_lky kblmZpby dh]^Z _jkbb \gmlj_gg_]h b \g_rg_]h ,3 aZ]heh\dh\ jZac

LjZgkihjlguc j_`bf

< ljZgkihjlghf j_`bf_ (63 aZsbsZ_l lhevdh ihe_agmx gZ]jmadm bkoh^gh]h
lhevdh hgZ ihf_sZ_lky Payload Data ijhlhdheZ (63 kf jbkmghd ljb wlhf \ j_ame
jmxs_f iZd_l_ bkihevam_lky bkoh^guc ,3 aZ]heh\hd k bkoh^gufb Z^j_kZfb gh
ghc ^ebghc k mdZaZgb_f gZ (63 dZd gZ ke_^mxsbc ijhlhdhe b k i_j_kqblZggh

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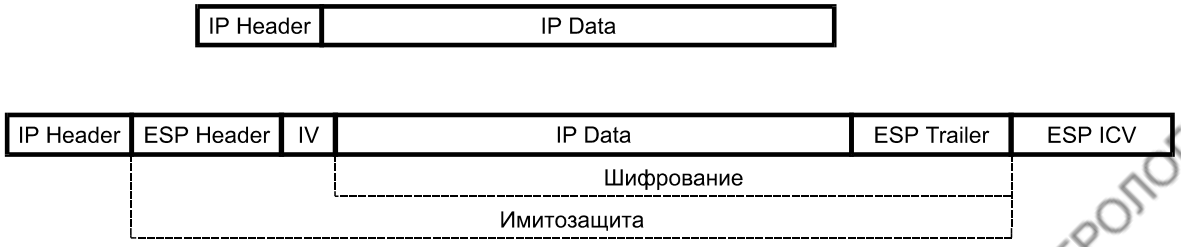
J ²

< wlhf kemqZ_ Next Header \ (63 iZd_l_ ^he`gh bf_lv agZq_gb_ kdhibjh\Zggh_ xs_]h ihey bkoh^gh]h ,3 aZ]heh\dZ < ljZgkihjlghf j_`bf_ ijhlhdhe (63 aZs bkoh^gh]h iZd_lZ gh g_ _]h ,3 aZ]heh\hd



Jbkmghd ² LjZgkihjlguc j_`bf bkoh^guc b aZsbs_gguc iZd_lu

ljb bkihevah\Zgbb ljZgkihjlgh]h j_`bfZ (63 k ijhlhdhehf ,3Y jZkrbj_gg lbiZ ©KRS E\ KRSª ©URXWLQJª b ©IUDJPHQWDWLRQª g_ ^he`gu aZsbsZ gu jZaf_sZlvky \ ,3 iZd_l_ ^h aZ]heh\dZ (63 JZkrbj_ggu_ aZ]heh\db lbiZ aZ\bkbfhklb hl `_eZ_fhc k_fZglbdb fh]ml dZd aZsbsZlvky ijhlhdhehf (63 lh ljb (63 iZd_lZ lZd b g_l < h[s_f kemqZ_ ij_^ihqlbl_evgh ihf_sZlv jZkrbj_ ©GHVWLQDWLRQ RSWLRQVª \ (63 iZd_l h[_ki_qb\Zy l_f kZfuf bo aZsblm

< ljZgkihjlghf j_`bf_ (63 g_ fh`_l ijbf_gylvky d hl^_evguf ,3 njZ]f_glZf qZ_ aZsblu njZ]f_glbjh\Zggh]h ,3 iZd_lZ i_j_^ ijbf_g_gb_f (63 ke_^m_l ba gh\blv bkoh^guc ,3 iZd_l

>hihegbl_evgh_ bgdZikmebjh\Zgb_ (63 iZd_lh\

ljhlhdhe (63 hij_^_e_g g_ihkj_^kl_ggh ih_jo ijhlhdheZ ,3 b \u]ey^bl dZ lhdhe ihe_ \ ,3 aZ]heh\d_ b^_glbnbpbjmxs__ ke_^mxsbc ijhlhdhe mdZau iehoh kh\f_klbfZ k mkljhckl\Zfb ljZgkeypbb k_l_\uo Z^j_kh\ 1$7 dhlhju_ iheZ]Zxlky gZ ijbkmlkl\b_ aZ]heh\dZ 7&3 beb 8'3 ihke_ aZ]heh\dZ ,3 b ij g_dhlhju_ ihey wlbo aZ]heh\dh\ ihjlu < kemqZ_ k (63 wlh g_\hafh`gh b 1 kl_ kemqZ_\ [ehdbjmxl ijhoh^_gb_ (63 iZd_lh\ Qlh[u h[_ki_qblv jZ[hlm ij 1$7 bkihevam_lky bgdZikmebjh\Zgb_ (63 iZd_lh\ \ 8'3 beb \ 7&3

BgdZikmebjh\Zgb_ (63 iZd_lh\ \ 8'3

ljb bgdZikmebjh\Zgbb (63 iZd_lZ \ 8'3 aZ]heh\hd 8'3 ihf_sZ_lky ,3 aZ]heh\dhf b aZ]heh\dhf (63 ljb wlhf ihe_ \ ,3 aZ]heh\d_ b^_glbnbpb lhdhe bf__l agZq_gb_ 8'3 Z ihjl gZagZq_gby 8'3 aZ]heh\dZ ^he`_g [ (63 iZd_lu ijboh^yl gZ lhl `_ ihjl qlh b ,.(Y iZd_lu ljbgbfZxsZy klhjhg bo ijhZgZebabjh\Z\ i_j\u_ [ZclZ ,.(Y iZd_lu ijboh^ysb_ gZ ihjl q_luj_o [Zcl gme_c \ lh \j_fy dZd i_j\u_ [ZclZ (63 aZ]heh\dZ kh^_j`Zl gbdh]^Z g_ jZ\gh gmex AZ]heh\hd 8'3 gbdZd g_ aZsbs_g b hl[jZku\Z_lky ihke_ mki_rghc h[jZ[hldb ihemq_ggh]h iZd_lZ GZ jbkmgd_ ihdZaZg nhj \Zgbb _]h \ 8'3

ljb bkihevah\Zgbb ljZgkihjlgh]h j_`bfZ kh\f_klgh k 8'3 bgdZikmeypb_ gZ imlb ke_^h\Zgby iZd_lZ \ ijbr_^r_f iZd_l_ agZq_gby ,3 Z^j_kh\ [m^ml \ke_^kl\b_ ljZgkeypbb k_l_\uo Z^j_kh\ lhkdhevdm ihe_ dhgljhevghc km]heh\dZo 7&3 b 8'3 \dexqZ_l \ k_[y agZq_gby k_l_\uo Z^j_kh\ ba ,3 aZ]h ^_dZikmeypbb gZ ijb_fghc klhjhg_ g_h[oh^bfh i_j_kqblZlv agZq_gb_ dh ihjlguo aZ]heh\dh\ lZdbf h[jZahf qlh[u hgZ [ueZ dhjj_dlghc Bkoh^gu_ a i_j_kq_lZ dhgljhevghc kmffu ijb_fgZy klhjhgZ ihemqZ_l \ fhf_gl kha^Zgb lhdheZ ,.(Y

BgdZikmebjh\Zgb_ (63 \ 8'3 kh]eZkh\u\Z_lky Z[hg_glZfb ihkj_^kl\hf ij kha^Zgbb (63 6$

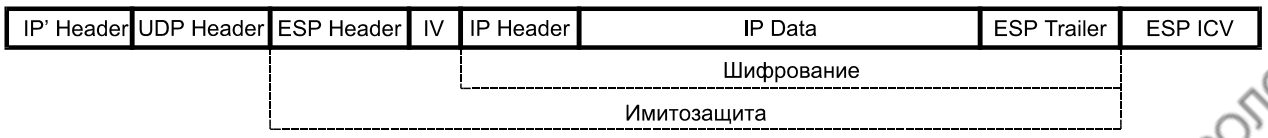
BgdZikmebjh\Zgb_ (63 iZd_lh\ \ 7&3

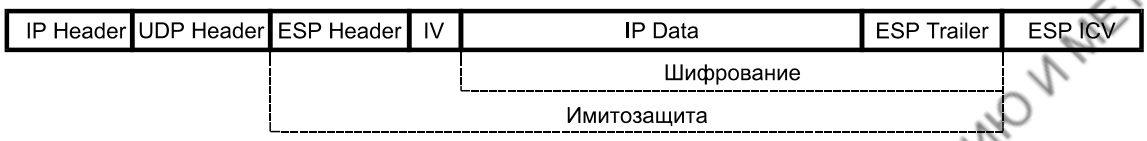
BgdZikmebjh\Zgb_ (63 iZd_lh\ \ 7&3 fh`_l [ulv bkihevah\Zgh dZd Zevl_ \Zgbx \ 8'3 \ l_o kemqZyo dh]^Z ijhlhdhe 8'3 aZ[ehdbjh\Zg gZ ijhf_`mlhc gZo lh kjZ\g_gbx k bgdZikmebjh\Zgb_f \ 8'3 bgdZikmebjh\Zgb_ \ 7&3 fh`

ФЕДЕРАЛЬНОЕ АГЕНТСТВО ПО ТЕХНИЧЕСКОМУ РЕГУЛИРОВАНИЮ И МЕТРОЛОГИИ

J ²

wnn_dlh\ \ qZklghklb fh`_l gZ[ex^Zlvky kms_kl\_ggh_ kgb`_gb_ kdhjhklb i_j_
kemqZyo dh]^Z aZsbsZ_fuf ijhlhdhehf lh`_ y\ey_lky 7&3 < k\yab k wlbf bgdZ
7&3 ^he`gh [ulv bkihevah\Zgh lhevdh \ l_o kemqZyo dh]^Z bkihevah\Zgb_ 8'3
guf BgdZikmebjh\Zgb_ (63 \ 7&3 kh]eZkh\u\Z_lky Z[hg_glZfb ihkj_^kl\hf ijhl
kha^Zgbb (63 6$

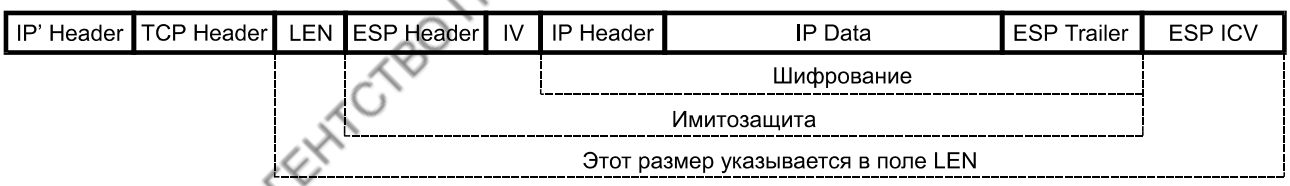


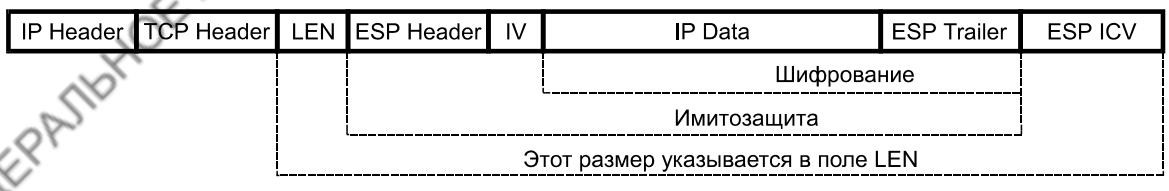


Jbkmghd ² BgdZikmebjh\Zgb_ (63 \ 8'3 \ lmgg_evghf b ljZgkihjlghf j_`bfZo

BgdZikmebjh\Zgb_ (63 \ 7&3 \uihegy_lky ZgZeh]bqgh bgdZikmebjh\Zgbx \ 8
ihf_sZ_lky f_`^m \g_rgbf ,3 aZ]heh\dhf b aZ]heh\dhf (63 ljb wlhf ihe_ \ ,3 aZ
nbpbjmxs__ ke_^mxsbc ijhlhdhe bf__l agZq_gb_ 7&3 Z ihjl gZagZq_gby 7
[ulv jZ_g AZf_lbf qlh \ wlhf kemqZ_ (63 iZd_lu ijboh^yl gZ lhl `_ ihjl ql
ljbgbfZxsZy klhjhgZ fh`_l jZaebqblv bo ijhZgZebabjh\Z\ i_j\u_ [ZclZ ,.(Y
^ysb_ gZ ihjl \k_]^Z gZqbgZxlky k q_luj_o gme_\uo [Zcl \ lh \j_fy dZd i_
aZ]heh\dZ kh^_j`Zl 63, agZq_gb_ dhlhjh]h gbdh]^Z g_ jZ\gh gmex AZ]heh\hd
b hl[jZku\Z_lky gZ ijbgbfZxs_c klhjhg_ ihke_ mki_rghc h[jZ[hldb ihemq_ggh]h
^_e_gby hl^_evguo (63 iZd_lh\ b ,.(Y iZd_lh\ \ ihlhd_ 7&3 i_j_^ dZ`^uf iZd_
ihe_LEN jZaf_jhf \ [ZclZ dhlhjh_ kh^_j`bl \ k_l_\hf ihjy^d_ [Zcl ^ebgm ke_^
(63 iZd_lZ beb ,.(Y iZd_lZ iexk jZaf_j kZfh]h ihey [ZclZ GZ jbkmgd_ ih
ijb bgdZikmebjh\Zgbb _]h \ 7&3

DZd b \ kemqZ_ k bgdZikmebjh\Zgb_f \ 8'3 ijb bkihevah\Zgbb 7&3 bgdZikm
j_`bf_ gZ ijb_fghc klhjhg_ g_h[oh^bfh dhjj_dlbjh\Zlv dhgljhevgmx kmffm \ aZ]h
ih^jh[g__ kf





Jbkmghd ² BgdZikmebjh\Zgb_ (63 \ 7&3 \ lmgg_evghf b ljZgkihjlghf j_`bfZo

ljbf_ qZ g bGZ jbkmgdZo ² ^ey mijhs_gby ihdZaZgh qlh ihe_ ,9 y\gh \dexqZ
aZsbsZ_fu_ bfblh\klZ\dhc < j_Zevghklb wlh aZ\bkbl hl bkihevam_fh]h ljZgknhjfZ < q
ljZgknhjfh\ j_Zebamxsbo rbnjh\Zgb_ ihe_ ,9 g_ \dexqZ_lky \ ^Zggu_ aZsbsZ_fu_ bfb
sblZ hl bkdZ`_gby hkms_kl\ey_lky g_y\guf h[jZahf >ey $($' ljZgknhjfh\ j_Zebamxsbo
lm ihe_ ,9 y\gh \dexqZ_lky \ ^Zggu_ aZsbsZ_fu_ bfblh\klZ\dhc Kf lZd`_

ФЕДЕРАЛЬНОЕ АГЕНТСТВО ПО ТЕХНИЧЕСКОМУ РЕГУЛИРОВАНИЮ И МЕТРОЛОГИИ

J ²

H[jZ[hldZ iZd_lh\

lheblbdZ [_ahiZkghklb

DZ`^uc ba Z[hg_glh\ ^he`_g bf_lv iheblbdm [_ahiZkghklb hij_^_eyxsm
nbd ^he`_g aZsbsZlvky b dZdbf h[jZahf lheblbdZ [_ahiZkghklb ij_^klZ\ey_
kibkhd ijZ\be dZ`^h_ ba dhlhjuo kh^_j`bl ke_^mxsmx bgnhjfZpbx

k_e_dlhj ljZnbdZ hij_^_ey_l dZdhc k_l_\hc ljZnbd ihiZ^Z_l ih^ ^_ckl
^_ckl\b_ k ljZnbdhf ih^iZ^Zxsbf ih^ k_e_dlhj ©ijhimklblvª ©hl[jhkbl
^bgZfbq_kdbc kibkhd aZsbs_gguo kh_^bg_gbc kha^Zgguo ^Zgguf ijZ\b
sblblvª

\hafh`gu_ j_`bfu bgdZikmeypbb ^ey ^_ckl\by ©aZsblblvª

kibkhd gZ[hjh\ ljZgknhjfh\ dhlhju_ fh]ml [ulv bkihevah\Zgu ^ey aZsbl
kl\by ©aZsblblvª

bgu_ iZjZf_lju g_ hlghkysb_ky d [_ahiZkghklb gZijbf_j mjh_gv `mjg

LZdhc kibkhd gZau\Z_lky 6HFXULW\ 3ROLF\ 'DWDEDVH 63'

AZsbs_ggh_ kh_^bg_gb_ 6$

DZ`^uc iZd_l h[jZ[Zlu\Z_lky \ g_dhlhjhf dhgl dkl_ gZau\Z_fhf aZsbs_
6HFXULW\ $VVRFLDWLRQ 6$ < ^Zgghf ^hdmf_gl_ jZkkfZljb\Zxlky lhevd
aZggu_ k ijhlhdhehf (63 ² (63 6$ AZsbs_ggu_ kh_^bg_gby kha^Zxlky b m
ijhlhdhehf kh]eZkh\Zgby dexq_c b kh^_j`Zl lZdb_ ^Zggu_ dZd

b^_glbnbdZlhj aZsbs_ggh]h kh_^bg_gby SPI

bkihevam_fu_ ljZgknhjfu

ljZgknhjf rbnjh\Zgby beb $($' Ze]hjfZ ENCR lbiZ

hipbhgZevgh ljZgknhjf bfblhaZsblu ESN lbiZ

l j b f _ q Z g ⇒ ^Zg²ghf ^hdmf_gl_ hibku\Z_lky bkihevah\Zgb_ lhevdh $($' Ze]h
nhjfZ lbiZ ENCR ihwlhfm ljZgknhjfZ INTEGZ _ bkihevam_lky

ljZgknhjf lbiZ ESN hij_^_eyxsbc bkihevah\Zgb_ jZkrbj_ggh]h ihjy^dh\h]h
ijbgbfZ_l agZq_gb_ ESN _keb bkihevah\Zgb_ (61 kh]eZkh\Zgh b QR(61 \ i

dexq ^ey ljZgknhjfZ rbnjh\Zgby HQRU

hipbhgZevgh dexq ^ey ljZgknhjfZ bfblhaZsblu

l j b f _ q Z g ⇒ ^Zg²ghf ^hdmf_gl_ hibku\Z_lky bkihevah\Zgb_ lhevdh $($' ljZgk
gby ihwlhfm dexq ^ey ljZgknhjfZ bfblhaZsblu hlkmlkl\m_l

k_e_dlhj ljZnbdZ hij_^_ey_l dZdhc ljZnbd ^he`_g aZsbsZlvky ^Zgguf
g_gb_f

ihjy^dh\uc ghf_j ke_^mxs_]h bkoh^ys_]h iZd_lZ 61 beb (61

kdhevays_ hdgh ghf_jh\ iZd_lh\ gZ ijb_fghc klhjhg_

j_`bf bgdZikmeypbb lmgg_evguc ljZgkihjlguc

lmgg_evguc ,3 Z^j_k ^ey lmgg_evgh]h j_`bfZ

bgu_ iZjZf_lju g_ hlghkysb_ky d [_ahiZkghklb gZijbf_j mjh_gv `mjg

ljhlhdhe (63 y\ey_lky h^ghgZijZ\e_gguf lh _klv h^gh aZsbs_ggh_ kh_^
aZsblm ljZnbdZ b^ms_]h lhevdh \ h^ghf gZijZ\e_gbb >ey aZsblu ^\mgZijZ
gh]h \ k_lb Bgl_jg_l Z[hg_glZf g_h[oh^bfh kha^Zlv ^\Z aZsbs_gguo kh_^b
\k_]^Z kha^Z_l (63 6$ ihiZjgh \g_ aZ\bkbfhklb hl lh]h [m^_l eb aZsbsZl
ljZnbd beb ^\mgZijZ\e_gguc

AZsbs_ggh_ kh_^bg_gb_ b^_glbnbpbjm_lky agZq_gb_f 63, 6HFXULW\ 3
(63 kf

DZ`^Zy ba klhjhg ojZgbl ^bgZfbq_kdmx lZ[ebpm Zdlb\guo aZsbs_gguo k
6$' 6HFXULW\ $VVRFLDWLRQ 'DWDEDVH

H[jZ[hldZ bkoh^ys_]h iZd_lZ

GZ \oh^ ijhlhdheZ (63 ihklmiZ_l ,3 iZd_l ljZ\beZ 63' ijhkfZljb\Zxlky hl
g_fm k p_evx ihbkdZ lZdh]h ba gbo k_e_dlhj ljZnbdZ dhlhjh]h \dexqZ_l k
?keb ijZ\beh g_ gZc^_gh beb kh^_j`bl ^_ckl\b_ ©hl[jhkblvª iZd_l mgbqlh
ijZ\beh kh^_j`bl ^_ckl\b_ ©ijhimklblvª iZd_l ijhimkdZ_lky

ФЕДЕРАЛЬНОЕ АГЕНТСТВО ПО ТЕХНИЧЕСКОМУ РЕГУЛИРОВАНИЮ И МЕТРОЛОГИИ

?keb gZc^_ggh_ ijZ\beh kh^_j`bl ^_ckl\b_ ©aZsblblvª lh ijhkfZljb\Z_lky l
kha^Zgguo ^Zgguf ijZ\behf k p_evx ihbkdZ m`_ mklZgh\e_ggh]h aZsbs_ggh]h
ljZnbdZ dhlhjh]h \dexqZ_l k_l_\u_ iZjZf_lju iZd_lZ ?keb lZdh\h]h g_ gZc^_gl
ijhlhdhem mijZ\e_gby dexqZfb ^ey kha^Zgby gh\h]h (63 6$ k l_fb iZjZf_ljZfb
ijZ\be_ Z kZf iZd_l klZ\blky \ ebkl h`b^Zgby H[jZ[hldZ lZdh]h iZd_lZ \hah[gh\
aZsbs_ggh_ kh_^bg_gb_ (63 [m^_l kha^Zgh

l j b f _ q Z g k ^Zgghf ^hdmf_gl_ ij_^iheZ]Z_lky bkihevah\Zgb_ ijhlhdheZ mijZ\e_
,.(Y ^ey kha^Zgby h^ghgZijZ\e_gguo XQLFDVW (63 6$

?keb ih^oh^ys__ (63 6$ gZc^_gh lh iZd_l aZsbsZ_lky \ khhl_lkl\bb k mdZ
jZf_ljZfb < qZklghklb
Z hkms_kl\ey_lky bgdZikmeypby iZd_lZ \ khhl_lkl\bb k mdZaZgguf \ (63
wlhf ijhbkoh^bl nhjfbjh\Zgb_ b ^h[Z\e_gb_ g_h[oh^bfuo aZ]heh\dh\ ² (63 b hip
,3 aZ]heh\dZ
[^Zggu_ iZd_lZ ^hihegyxlky hipbhgZevguf TFC Padding b h[yaZ_lky Padding
g_gbyfb aZihegyxlky ihey Padding Next Header
\ \uqbkey_lky ghf_j beb jZkrbj_gguc ghf_j ^ey ^Zggh]h iZd_lZ iml_f m_
pm _]h agZq_gby ih hlghr_gbx d ghf_jm ij_^u^ms_]h hlijZ\e_ggh]h iZd_lZ
] aZihegyxlky ihey (63 aZ]heh\dZ SPI b
^ \uqbkey_lky _dlhj bgbpbZebaZpbb ^ey ^Zggh]h iZd_lZ \ khhl_lkl\bb k m
lh]jZnbq_kdbf ij_h[jZah\Zgb_f _]h agZq_gb_ aZibku\Z_lky \ ihe_ IV
_ hij_^_ey_lky dexq aZsblu iZd_lZ ijb g_h[oh^bfhklb bgbpbbjm_lky ijhp_k
q_c kf
` iZd_l aZrbnjh\u\Z_lky b bfblhaZsbsZ_lky k bkihevah\Zgb_ mdZ aZgguf
\ (63 6$ dexqZ aZsblu iZd_lZ b \uqbke_ggh]h _dlhjZ b ESP
_lky d iZd_lm

l j b f _ q Z g k ^Zgghf ^hdmf_gl_ hibku\Z_lky bkihevah\Zgb_ lhevdh $($' Ze]hjblfh
nhjfZ lbEZ ihwlhfm bfblhaZsblZ ^Zgguo iZd_lZ hkms_kl\ey_lky \f_kl_ k bo rbnjh\Zgb_

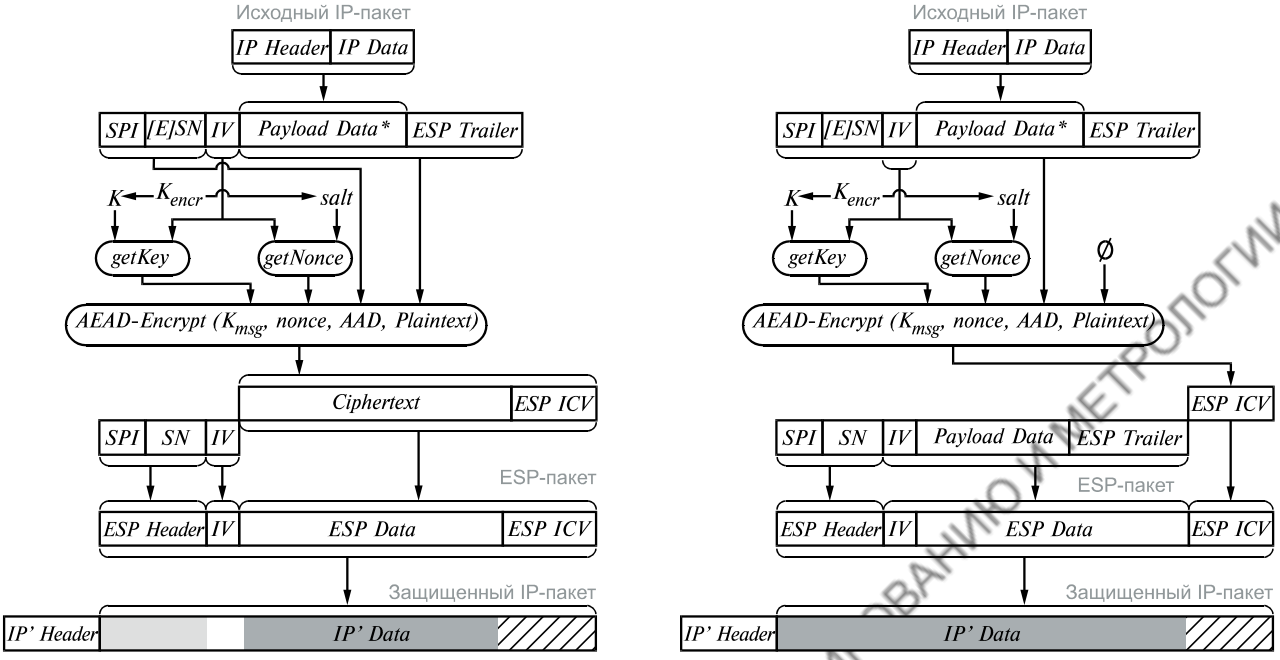
b hipbhgZevgh iZd_l bgdZikmebjm_lky \ 8'3 beb 7&3 \ khhl_lkl\bb k iZjZf_
d ijhbkoh^bl h[gh\e_gb_ ^bgZfbq_kdbo ihe_c (63 6$ lZdbo dZd
ihjy^dh\uc ghf_j beb jZkrbj_gguc ghf_j ^Zggh]h iZd_lZ dZd ihke_^g_]h h
dhebq_kl\h aZsbs_gguo iZd_lh\
h[t_f aZsbs_gguo ^Zgguo
,9 ijb g_h[oh^bfhklb
ijhqb_ ^bgZfbq_kdb_ ihey 6$

l j b f _ q Z g b ^Zgguc Ze]hjblf ij_^iheZ]Z_l qlh h[jZ[hldZ iZd_lh\ ijhbkoh^bl \ h
< kemqZ_ fgh]hihlhqghc h[jZ[hldb iZd_lh\ \ jZfdZo h^gh]h aZsbs_ggh]h kh_^bg_gby g_h
gbl_evgu_ f_ju ^ey kbgojhgbaZpbb ijh_jdb b baf_g_gby h[sbo ^Zgguo 6$ f_`^m ihlhdZf

GZ jbkmgd_ ihdZaZgZ h[jZ[hldZ bkoh^ys_]h iZd_lZ \ kemqZ_ bkihevah\Zg
fZ Z gZ jbkmgd_ ² \ kemqZ_ bkihevah\Zgby ljZgkihjlgh]h j_`bfZ
ljb g_h[oh^bfhklb j_amevlbjmxsbc iZd_l i_j_^ hlijZ\dhc fh`_l njZ]f_glbjh\Z
f_j _keb _]h jZaf_j ij_\ukbe fZdkbfZevguc ebfbl jZaf_jh\ iZd_lh\ ^ey \uoh^gh]h
n_ckZ

ФЕДЕРАЛЬНОЕ АГЕНТСТВО ПО ТЕХНИЧЕСКОМУ РЕГУЛИРОВАНИЮ И МЕТРОЛОГИИ

J ²



а) Есть шифрование б) Нет шифрования

* ² \ aZ\bkbfhklb hl bkihevah\Zgby ^hihegbl_evgh]h aZihegh_gby \ CRC Padding
eb[h \ lhqghklb ,jZ \gh FNHW eb[h ij_^klZ\ey_l kh[hc dhgdZl_gZpbx ,3 'DWD C

² bfblhaZsbs_ggu_ ^Zggu_

² bfblhaZsbs_ggu_ b aZrbnjh\Zggu_ ^Zggu_

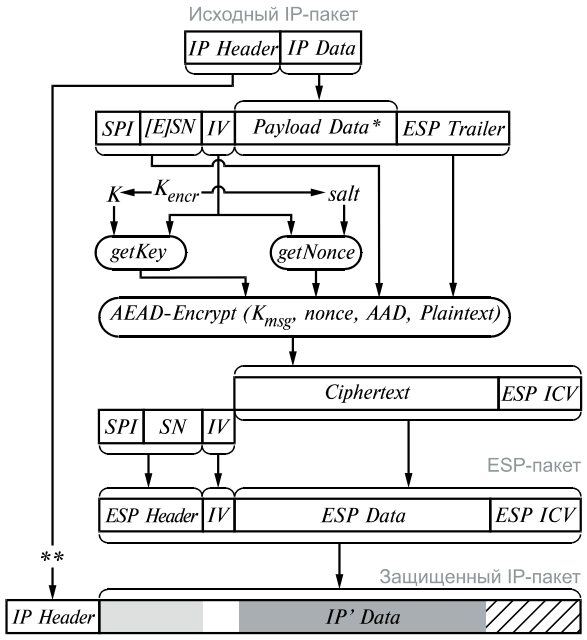
² agZq_gb_ bfblh\klZ\db

Jbkmghd ² H[jZ[hldZ bkoh^ys_]h iZd_lZ \ kemqZ_ bkihevah\Zgby lmgg_ev

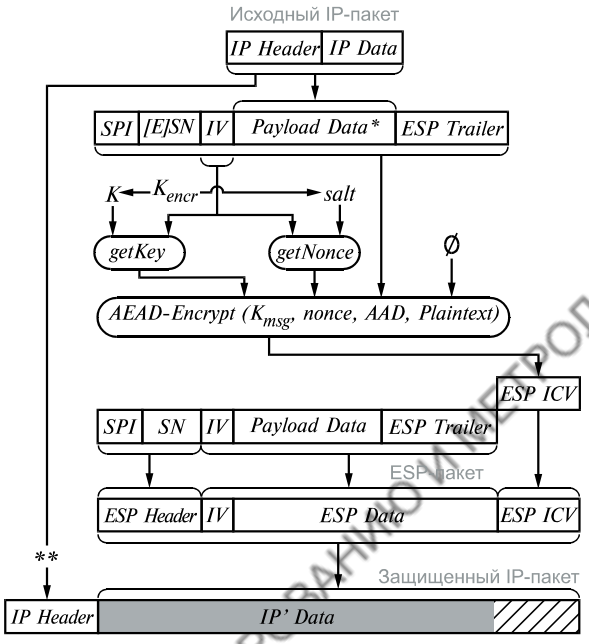
l j b f _ q Z g b y
Nmgdjbb JHW . HJ\HW 1 RaZ\hbkyl hl bkihevam_fh]h ljZgknhjfZ ENCR
Nhjfbjh\Zgb_ $$' \ h[s_f kemqZ_ aZ\bkbl hl bkihevam_fh]h ljZgknhjfZ ENCR

ФЕДЕРАЛЬНОЕ АГЕНТСТВО ПО ТЕХНИЧЕСКОМУ РЕГУЛИРОВАНИЮ И МЕТРОЛОГИИ

J ²



а) Есть шифрование



б) Нет шифрования

* ² \ aZ\bkbfhklb hl bkihevah\Zgby ^hihegbl_evgh]h aZihegbl_evgh]h (Payload Data Padding)
eb[h \ lhqghklb jZ\gh IP-Data ij_^klZ\ey_l kh[hc dhgdZl_gZpbx ,3 'DWD G 7)& 3DG

** ² \ ljZgkihjlghf j_`bf_ jZ[hlu ijhlhdheZ ,3 aZ]heh\hd iZd_lZ hklZ_lky g_baf_gguf k l
baf_g_gby iZjZf_ljZ kh^_j`Zs_]h ^ebgm iZd_lZ k mdZaZgb_f gZ (63 dZd gZ ke_
b k i_j_kqblZgghc dhgljhevghc kmffhc

² bfblhaZsbs_ggu_ ^Zggu_

² bfblhaZsbs_ggu_ b aZrbnjh\Zggu_ ^Zggu_

² agZq_gb_ bfblh\klZ\db

Jbkmghd ² H[jZ[hldZ bkoh^ys_]h iZd_lZ \ kemqZ_ bkihevah\Zgby ljZgkihjlgh]h

FZl_fZlbq_kdh_ hibkZgb_ ijhp_kkZ nhjfbjh\Zgby (63 iZd_lZ
Bkoh^guc iZd_l ,3 3DFNHW ij_^klZ\ey_lky \ \b^_

,3 3DFNHW ,3 +HDGHU G ,3 'DWD

]^_ ,3 +HDGHU ² aZ]heh\hd bkoh^gh]h ,3 iZd_lZ
,3 'DWD ² ihe_agZy gZ]jmadZ bkoh^gh]h ,3 iZd_lZ
Nhjfbjm_63 +HDGHU
(63 +HDGHU 63, G 61

]^_ 63, ² iheZ]Z_lky jZ\guf agZq_gbx 63, ^Zggh]h aZsbs_ggh]h kh_^bg_gby
61 ² \uqbkey_lky iml_f m_ebq_gby gZ _^bgbpm ghf_jZ ij_^u^ms_]h iZd_lZ
^Zgghfm aZsbs_gghfm kh_^bg_gbx
Nhjfbjm_9 ky
?keb (63 6$ ih^jZamf_\Z_l bkihevah\Zgb_ _dlhjZ bgbpbZebaZpbb lh agZq_
\ khhl_lkl\bb k lj_[h\Zgbyfb ljZgkihjlgh]h \ oh^_ kha^Zgby (63 6
ih^jh[g__
Nhjfbjm_63 'DWD (63 ,&9
Z Nhjfbjmxlky ihey 3D\OrDG 'DWD (63 G 7UDW O HU
?keb \ (63 6$ hij_^_e_g lmgg_evguc j_`bf jZ[hlu ijhlhdeZ (63 Dlhhjfbjm_l
ky ke_^mxsbf h[jZahf
3D\ORDG 'DWD ,3 +HDGHU G ,3 'DWD G 7)& 3DGGLQJ

]^_ 7)& 3DGGLQJ • %

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J ²

?keb \ (63 6$ hij_^_e_g ljZgkihjlguc j_`bf jZ[hlu ijZ[hlu ke_^mxsbf h[jZahf 3D\ORDG 'DWD ,3 'DWD G 7)& 3DGGLQJ

]^_ 7)& 3DGGLQJ • %
[(63 7UDLOHU 3DGGLQJ G 3DG /HQJWK G 1H[W +HDGHU

]^_ 3DGGLQJ G 3DG /HQJWK G 1H[W +HDGHU \ kemqZ_ bkihevah\Zgb_ ^jm]bo kihkh[h\ nhjfbjh\Zgby ^hiheg_gby i
jZaf_jm [ehdZ bkihevam_fh]h ljZgkihjlguc j_`bf jZ[hlu ijZ[hlu ke_^mxsbf h[jZahf ENCR kf
gby j_dhf_g^m_lky \u[bjZlv fbgbfZevguc jZaf_j ^hiheg_gby m^h\e_l\hjyxs

l j b f _ q Z g b > Zgg ² uc Ze]hjblf y\ey_lky j_dhf_g^m_fuf kihkh[hfZ ^hiheg_gby
h^gZdh > @ ^himkdZ_l bkihevah\Zgb_ ^jm]bo kihkh[h\ nhjfbjh\Zgby ^hiheg_gby i
hl[jZku\Zlv iZd_l _keb ^hiheg_gb_ knhjfbjh\Zgh bguf kihkh[hf

3DG /HQJWK j`bl ^eb 3DG GLQJ
1H[W +HDGHU \ kemqZ_ bkihevah\Zgb_ ^jm]bo kihkh[h\ nhjfbjh\Zgby ^hiheg_gby i
\ kemqZ_ ,3Y ^ey ljZgkihjlgh]h j_`bfZ agZq_gb_ dhibjm_lky ba ihey ^ey
bkoh^gh]h ,3 aZ]heh\dZ ijb nhjfbjh\Zgbb imkl[hhW G +HDGHU
\ lZd_l aZsbsZ_lky k bkihevah\Zgb_f iZjZf_ljh\ mdZaZgguo \ (63 6$ \
qZxlky agZq_gb_ 3D\ORDG 'DWD

&LSKHUWH[W G (63 ,&9 $(,$QRQFH\SW$'. 3ODLQWH[W

]^_ Ze]hjblf Zml_glbnbpbjh\Zggh]h $(,$QRQFH\SW aZ^Z_lky kh]eZkh\Zgguf
fhf ENCR kf

l j b f _ q Z g b > jZf ² Zo ^Zggh]h ^hdmf_glZ ij_^iheZ]Z_lky qlh \ oh^_ \ujZ[hldb
^ey ljZgknhjfZ ENCR kh]eZkh\Zg $(,$QRQFH\SW Ze]hjblf

agZq_gb_ 3D\ORDG 'DWD bQRQFH\Z[Zlu\Zxlky ba kexqZ _dlhjZ bgbpbZe 9 aZ]hbl_l
kl\bb k Ze]hjblfZfb hij_^_e_ggufb \ ljZgknhjf_ ENCR kf b
agZq_gb_ $(,$QRQFH\SW nhjfbjm_lky \ aZ\bkbfhklb hl kh]eZkh\Zgguf ENCR kf
l eb ljZgknhjf ENCR bkihevah\Zgb rbnjh\Zgby kf lZ[ebpm
agZq_gb_ 3D\ORDG 'DWD nhjfbjm_lky \ aZ\bkbfhklb hl lh]h kh]eZkh\Zgguf ENCR kf
gb_ rbnjh\Zgby kf lZ[ebpm

L Z [e b p Z ² NhjfZl 3ODLQWH[W

Rbnjh\Zgb_	3ODLQWH[W
?klv rbnjh\Zgb_	3D\ORDG 'DWD 7UDLOHU
G_l rbnjh\Zgby	

] J_amevlbjmxs__ agZq_gb_ nhjfbjm_lky \ aZ\bkbfhklb hl lh]h kh]eZkh\Zgguf
nhjf ENCR bkihevah\Zgb_ rbnjh\Zgby kf lZ[ebpm

L Z [e b p Z ² NhjfZl (63 'DWD

Rbnjh\Zgb_	(63 'DWD
?klv rbnjh\Zgb_	&LSKHUWH[W
G_l rbnjh\Zgby	3D\ORDG 'DWD 7UDLOHU

J_amevlbjmxsbc iZd_l ,3¶ 3DFNHW nhjfbjm_lky ke_^mxsbf h[jZah

,3¶ 3DFNHW ,3¶ +HDGHU G (63 +HDGHU G ,9 G (63 'DW

]^_ ,3¶ +HDGHU gh\uc ,3 aZ]heh\hd k gh\ufb agZq_gbyfb ,3 Z^j_kh\ h[uqg
oh^ghf ,3 aZ]heh\hd +HDGHU _keb \ (63 6$ hij_^_e_g lmgg_evguc

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J ²

ijhlhdheZ eb[h bkoh^guc,3 iZd_lZ]hey (63 iZd_lZ dhjj_dlbjh\Zgghc ^ebghc k mdZ
gb_f gZ (63 dZd gZ ke_^mxsbc ijhlhdhe b k i_j_kqblZgghc dhgljhevg
\ (63 6$ hij_^_e_g ljZgkihjlguc j_`bf jZ[hlu ijhlhdheZ
(63 +HDG,9U (63 'DW(D3 ,&9 ihey (63 iZd_lZ kf nhjfbjh\Zgb_ dhlhju
hibkZgh \ ²

H[jZ[hldZ \oh^ys_]h iZd_lZ

?keb ,3 iZd_l ijb i_j_kued_ [ue jZa[bl gZ njZ]f_glu lh i_j\hgZqZevgh \uih
gh\e_gb_ p_eh]h ,3 iZd_lZ ba ihemq_gguo ,3 njZ]f_glh\

Ijb ihemq_gbb (63 iZd_lZ Z^j_kh\Zggh]h ^Zgghfm Z[hg_glm hg h[jZ[Zlu\
h[jZahf

Z ?keb iZd_l [ue bgdZikmebjh\Zg \ 8'3 beb \ 7&3 lh hkms_kl\ey_lky _]h
[jZku\Zgb_f wlbo aZ]heh\dh\

[Ba ihemq_ggh]h ,3 iZd_lZ \u^_ey_lky (63 iZd_l b \uihegy_lky ihbkd 6$ \
63, ba (63 aZ]heh\dZ _keb aZsbs_ggh_ kh_^bg_gb_ g_ gZc^_gh iZd_l hl[jZku

I j b f _ q Z g b ^Zgg²u_ ki_pbnbdZpbb g_ j_]eZf_glbjmxl dhgdj_lguc f_oZgbaf ihbk
\hafh`guo \ZjbZglh\ y\ey_lky ihbkd ih gZb[he__ iheghfm khhl_lkl\bx ljbZ^_ ,3 Z^j_k
gZagZq_gby 63, lh _klv kgZqZeZ aZsbs_ggh_ kh_^bg_gb_ bs_lky ih \k_f lj_f djbl_jbyf
^\mf ihke_^gbf _keb g_ gZc^_gh lh lhevdh ih 63,

\ Ghf_j ihemq_ggh]h (63 iZd_lZ ijh_jy_lky k bkihevah\Zgb_f kdhevays_]h
iZd_l b^_glbnbpbjh\Zg dZd ih\lhjguc beb mklZj_lrbc hg hl[jZku\Z_lky

] Hij_^_ey_lky dexq aZsblu iZd_lZ ijb g_h[oh^bfhklb bgbpbbjm_lky ijhp_kl
q_c kf

^ Ijh\h^blky ijh_jdZ ,&9 iZd_lZ _keb hgZ ijhreZ mki_rgh lh \uihegy_lky
iZd_lZ k bkihevah\Zgb_f lEXCR nhjfZ aZgg]h \ (63 6$ dexqZ aZsblu iZd_lZ b _
pbZebaZpbb ba iZd_lZ bgZq_ iZd_l hl[jZku\Z_lky

I j b f _ q Z g b ^Zgg²ghf ^hdmf_gl_ hibku\Z_lky bkihevah\Zgb_ lhevdh $($' Ze]hjblfh
nhjfZ lbEXCR ihwlhfm jZkrbnjh\Zgb_ ^Zgguo ijhbkoh^bl iZjZee_evgh k ijh_jdhc ,&9 iZd

_ Hl[jZku\Zxlky h[yaZ_kdmgh b hipbhgZevgh_ TFC Padding Adding eg_gby

` Bkoh^guc iZd_l j_dhgkljmbjm_lky \ khhl\_lkl\bb k j_`bfhf bgdZikmeypbb
(63 6$ k bkihevah\Zgb_f agZq_gbyfb Zex_kdmgh ihey_keb j_`bf bgdZikmeypbb iZd_lZ g_ kh
^Z_l k j_`bfhf bgdZikmeypbb mdZaZgguf \ (63 6$ iZd_l hl[jZku\Z_lky dZd gZ
[_ahiZkghklb

b K_l_\u_ iZjZf_lju ihemq_ggh]h iZd_lZ ijh_jyxlky gZ \oh`^_gb_ \ k_e_dl
6$ _keb hgb g_ \oh^yl lh iZd_l hl[jZku\Z_lky dZd gZjmrZxsbc iheblbdm [_ahi

d Ijhbkoh^bl h[gh\e_gb_ ^bgZfbq_kdbo ihe_c 6$ lZdbo dZd
]jZgbpu b beb kh^_j`bfh_ kdhevays_]h hdgZ ghf_jh\ iZd_lh\
dhebq_kl\h h[jZ[hlZgguo \oh^ysbo iZd_lh\
h[t_f h[jZ[hlZgguo ^Zgguo
ijhqb_ ^bgZfbq_kdb_ ihey 6$

e ?keb ihemq_gguc iZd_l [ue bgdZikmebjh\Zg \ 8'3 beb 7&3 b Z^j_k be
g_ kh\iZ^Zxl k mdZaZggufb \ 6$ agZq_gbyfb b 6$ ^himkdZ_l h[gh\e_gb_ lmgg_evg
guc Z^j_k b ihjl 7&3 8'3 bgdZikmeypbb h[gh\eyxlky agZq_gbyfb Z^j_kZ b ihjlZ bk
lhemq_gguc iZd_l hlijZ\ey_lky \ 7&3 ,3 kl_d kbkl_fu

I j b f _ q Z g b ^Zgg²uc Ze]hjblf ij_^iheZ]Z_l qlh h[jZ[hldZ iZd_lh\ ijhbkoh^bl \ h
< kemqZ_ fgh]hihlhqghc h[jZ[hldb iZd_lh\ \ jZfdZo h^gh]h aZsbs_ggh]h kh_^bg_gby g_h
gbl_evgu_ f_ju ^ey kbgojhgbaZpbb ijh_jdb b baf_g_gby h[sbo ^Zgguo 6$ f_`^m ihlhdZf

FZl_fZlbq_kdh_ hibkZgb_ ijhp_kkZ ba\e_q_gby ^Zgguo ba (63 iZd_lZ
Ijh\h^blky ijh_jdZ ,&9 ihemq_ggh]h iZd_lZ b _]h jZkrbnjh\Zgb_ ?k
g_ ijhreZ lh iZd_l hl[jZku\Z_lky

3ODLQWH[W $($'HOHPHQWH &LSKHUWH[W ,&9

]^_ Ze]hjblf jZkrbnjh\Zgby b ijh_jdb b f(($'HOHPHQWH aZ^Z_lky kh]eZkh\Zgguf ljZ
nhjfhfENCR kf

agZq_gbyfb QRQuFZ[Zlu\Zxlky ba Kexqb_dlhjZ bgbpbZe9 aZkhhl_lkl\bb
k Ze]hjblfZfb kh]eZkh\Zggufb lEXCR nhjfhf b

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J ²

agZq_gb_ $$' nhjfbjm_lky \ aZ\bkbfhklb hl kh]eZkh\Zgby ENCR_KUZNYECHIK_MGM_KTREE ENCR_MAGMA_MGM_KTREE
eb ljZgknhjf ENCR bkihevah\Zgb_ rbnjh\Zgby kf lZ[ebpm
agZq_gb_ &LSKHUWH[W bkihevah\Zgb_ rbnjh\Zgby kf lZ[ebpm ENCR_KUZNYECHIK_MGM_MAC_KTREE ENCR_MAGMA_MGM_MAC_KTREE

L Z [e b p Z ² NhjfZl &LSKHUWH[W

Rbnjh\Zgb_	&LSKHUWH[W
?klv rbnjh\Zgb_	(63 'DWD
G_l rbnjh\Zgby	

?keb ihe_ 1H[W (63 7UDLQLQJ agZq_gb_ iZd_l hl[jZku\Z_lky
Hl ihemq_ggh]h agZq_gby 1H[W ijb hlkmlkl\bb rbnjh\Zgby (63 'DWD ²hl[jZku
\Z_lky (63 7UDLQLQJ agZq_gb_ 3D\ORDG 'DWD
Hl agZq_gby 3D\ORDG 'DWD iZd_u\Z_lky & 3DGGLQJ gZebqbb
Ihemq_ggZy _ebqbgZ bgl_jij3lbjm_lky iZdlhf
Z _keb (63 6$ bkihevam_l lmgg_evguc j_`bf lh ,3 'DWD ,3 3DFNHW nhjfbjm_l
[_keb (63 6$ bkihevam_l ljZgkihjlguc j_`bf lh ,3 3DFNHW nhjfbjm_lky

,3 3DFNHW ,3 +HDGHU G ,3 'DWD

]^_,3 +HDGHU G iZ]heh\hd bkoh^gh]h ,3 iZd_lZ \dhlhjhf kdhjj_dlbjh\ZgZ ^ebgZ
s__ ljZgkihjlguc ijhlhdhe aZf g_gh gZ agZq_gb_ ihey 1H[W +H

Bkihevah\Zgb_ jhkkbckdbo djbilh]jZnbq_kdbo Ze]hjblfh\ \ ijh

Kh]eZkh\Zgb_ iZjZf_ljh\ (63 6$

< khhl_lkl\bb k gZklhysbfb j_dhf_g^Zpbyfb \ ijhp_kk_ kh]eZkh\Zgby i
ijhlhdhe_ ,.(Y ^ey ljZgknhjf ENCR_KUZNYECHIK_MGM_KTREE ^m_l bkihevah\Zlv lhevdh iZjZf_lju ij_^k
\ lhkdhevdm ^ey ljZgknhjf ENCR_MAGMA_MGM_KTREE bkihevamxlky $($' Ze]hjblfu ijb kh]eZkh\Z
f_ljh\ 6$ ljZgknhjf ENCR_KUZNYECHIK_MGM_MAC_KTREE g_ ^he`_g bkihevah\Zlvky >Zggu_ j_dhf_g^Zpbb g
h]jZgbq_gbc gZ kh]eZkh\Zgb_ ijhqbo iZjZf_ljh\ (63 6$

LjZgknhjfu

GZklhysb_ j_dhf_g^Zpbb hij_^_eyxl q_luj_ ljZgknhjfZ lbiZ
ENCR_KUZNYECHIK_MGM_KTREE
ENCR_MAGMA_MGM_KTREE
ENCR_KUZNYECHIK_MGM_MAC_KTREE
ENCR_MAGMA_MGM_MAC_KTREE
<k_ hibkZggu_ ljZgknhjfu ij_^klZ\eyxl kh[hc $($' Ze]hjblfu k ^b_jkb
q_c LjZgknhjfu [Zabjmxlky gZ Ze]hjblfZo ©Dmag_qbdª ENCR_KUZNYECHIK_MG
KUZNYECHIK_MGM_MAC_KTREE © FZ]fZª ENCR_MAGMA_MGM_KTREE MAGMA_MGM_MAC_KTREE
hibkZggu o \ =HKL J ² jZa^_eu b :e]hjblfu bkihevamxlky \ j_`bf
J ²
LjZgknhjfu ENCR_KUZNYECHIK_MGM_KTREE ENCR_MAGMA_MGM_KTREE [_ki_qb\Zxl dhgnb
^_gpbZevghklv b bfblhaZsblm i_j_^Z\Z_fuo ^Zggu o iml_f Zml_glbnbpbjh
ljZgknhjfu ENCR_KUZNYECHIK_MGM_MAC_KTREE ENCR_MAGMA_MGM_MAC_KTREE [_ki_qb\Zxl
lhevdh bfblhaZsblm l_ rbnjh\Zgb_ g_ ijbf_gy_lky

I j b f _ q Z g b _ ² Bkihevah\Zgb_ ENCR_KUZNYECHIK_MGM_KTREE $(' Ze]hjblfhf ^ey h[_ki_q_gby k\h
bfblhaZsblu [_a dhgnb^_gpbZevghklb \f_klh ENCR_KUZNYECHIK_MGM_KTREE ^m_l gh l_f qlh \ ijhlhdhe_
lhevdh ljZgknhjf ENCR_KUZNYECHIK_MGM_KTREE Zfh`_l bf_lv _dlhj bgbpbZebaZpbb lh _klv lhevdh ljZgknhjf
ihkljhblv ^_j_\h dexq_c h[_ki_qb\Zxs__ bo ^b_jkbnbdZpbx

< ijhlhdhe_ ,.(Y ijb kh]eZkh\Zgbb iZjZf_ljh\ ENCR_KUZNYECHIK_MGM_
KTREE b ENCR_KUZNYECHIK_MGM_MAC_KTREE ^klZ\e_gu dZd $($' Ze]hjblfu k ^eb qhc de
[blZ ba dhlhjuo i_j\u_ [bl bkihevamxlky dZd dhjg_\hc dexq 6$ K Z
\ dZq_kl_ kheb VDOW < k\hx hq ENCR_KUZNYECHIK_MGM_KTREE ENCR_MAGMA_MGM_KTREE ENCR_MAGMA_MGM_

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J ²

MAC_KTREEij_^klZ\e_gu dZd $($' Ze]hjblfu k ^ebghc dexqZ [bl ba dhlhjuo i_
bkihevamxlky dZd dhjg_\hc dexq 6$ K Z ihke_^mxsb_ [blZ ² \ dZq_kl_ k
ljZgknhjfh\ ^ebgu dexq_c y\eyxlky nbdkbjh\Zggufb ihwlhfm ijb bo kh]eZkh\Zgb
Key Length g_ ^he`_g bkihevah\Zlvky

>Zggu_ ljZgknhjfu g_ gZdeZ^u\Zxl lj_[h\Zgbc gZ \ujZ\gb\Zgb_ ^ebgu aZsbs
kf

;ehqguc rbnj

Hij_^_e_ggu_ \ gZklhysbo j_dhf_g^Zpbb ENCRjZgkZhjfkl_ [ehqgh]h rbnjZ bk
ihevamxl rbnju ©FZ]fZª b ©Dmag_qbdª hij_^_e_ggu_ \ =HKL J [ehqgh]h rbnjZ
[ehdZ khklZ\ey_l [bl ^ey rbnjZ ©Dmag_qbdª b [blZ ^ey rbnjZ ©FZ]fZª ^eb
kemqZyo khklZ\ey_l [bl

L Z [e b p Z ² :e]hjblfu [ehqgh]h rbnjh\Zgby

LjZgknhjf ENCR	;ehqguc rbnj
ENCR_KUZNYECHIK_MGM_KTREE	©Dmag_qbdª =HKL J ² jZa^_e
ENCR_MAGMA_MGM_KTREE	©FZ]fZª =HKL J ² jZa^_e
ENCR_KUZNYECHIK_MGM_MAC_KTREE	©Dmag_qbdª =HKL J ² jZa^_e
ENCR_MAGMA_MGM_MAC_KTREE	©FZ]fZª =HKL J ² jZa^_e

$($' Ze]hjblf

Hij_^_e_ggu_ \ gZklhysbo j_dhf_g^Zpbb ENCRjZgkZhjfkl_ $($' Ze]hjblfZ bk
ihevamxl [ehqguc rbnj \ j_`bf_ 0*0 hibkZgghf \ J ² ENCR_ ljZgknhjf
KUZNYECHIK_MGM_KTREE ENCR_KUZNYECHIK_MGM_MAC_KTREE bgZ bfblh\klZ\ihe Z]Z_lky
jZ\ghc [blZf ^ey ljZgknhjfu ENCR_MAGMA_MGM_KTREE ENCR_MAGMA_MGM_MAC_KTREE Eihev
am_lky ihegZy bfblh\klZ\dZ l_ 6

Nmgdpby aZrbnjh\Zgb(QFU\SW aZ^Z_lky \ khhl_lkl\bb kh ke_^mxs_c nhjfm

$($' (QFU\SW QRQFH $$' 3ODLQWH[W &LSKHUWH[W G ,&
PVJ

]^_ 0*0QRQFH $$' &LSKHUWH[W ,&9 0*0QRQFH $$' 3ODLQWH[W
PVJ

0*0QRQFH QRQ@H [I G QRQFH> _QRQFH_@

Nmgdpby jZkrbnjh\Zgb'HFU\SW aZ^Z_lky \ khhl_lkl\bb kh ke_^mxs_c nhjfm

$($' 'HFU\SW QRQFH $$' &LSKHUWH[W UHV
PVJ

]^_ UHV¶ 0*0 'HFU\SW QRQFH $$' &LSKHUWH[W ,&9
PVJ

0*0QRQFH QRQ@H [I G QRQFH> _QRQFH_@

$$\begin{cases} res = & \textit{Plaintext} \text{ в случае, если } res' = (AAD, Plaintext), \\ & \text{в случае, если } res' \text{ содержит ошибку.} \end{cases}$$

NhjfZl dexqZ ljZgknhjfZ

>ebgZ dexqZ ljZgknhjfZ hkl Z\ey_l [blZ NOHQ ^ey ENCR_ ljZgknhjfh\
KUZNYECHIK_MGM_KTREE ENCR_KUZNYECHIK_MGM_MAC_KTREE [bl NOHQ ^ey ljZgknhj
fh\ ENCR_MAGMA_MGM_KTREE ENCR_MAGMA_MGM_MAC_KTREE.

GZklhysb_ j_dhf_g^Zpbb ij_^iheZ]Zxl qlh ba kl e xqZ hjfZ jZgknhjfZ dhjg_\hc
dexq . 32%b kljhdZ VDOW 32%
NOHQ HQFU
K G VDOW HQFU .

:e]hjblf ^b_jkbnbdZpbb dexqZ

Dexq aZsblu khhl_lkl\bb kh ke_^mxsbf h[jZahf
PVJ

i_j_^Z\Z_fuo \ _dlhj_ bgbpbZebaZpbb ke_^mxsbf h[jZahf

ESPTREE . L L L 'LYHUV 'LYHUV 'LYHUV O67 5 O67 5i
1 2 3 1 2 3 1 2 3

]^_ iB i LB ² iZjZf_lju ^b_jkbnbdZpbb dexqZ
1 2 3 2

O O O ² dhgklZglgu_ $6&,, kljhdb ^ebghc [Zcl [_a gme_\hl \ dhgp_ klj
1 2 3 1 level1 2 level2 3 level3

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J ²

'LYHUV . /Ze]h²jblf ^b_jkbnbdZpbb Kb Bexqh Z^Zgguf ^b_jkbnbdZpbb f_ld_
L B dhlhjuc aZ^Z_lky k ihfhsvx Ze]hjblfZ .')B*2675 B B
kZggh]h \ J ² ih^jZ aZ_ehU/ ' .') LQ ODEHO VHH
NhjfZl _dlhjZ bgbpbZebaZpbb khh[s_gby
DZ`^h_ aZsbs_ggh_ khh[s_gb_ kh^_j`bl _dlhj bgbpbZebaZpbb <_dlhj klZ\ey_l kh[hc dhgdZl_gZpbx q_luj_o _ebqbg

$$,9=i\ G_2,G_3,G\ 3180$$
$$_2=STR_2\ _2\ L$$
$$_3=STR_2\ _3\ L$$
$$3180=STR_3\ SQXP$$

]^_ iB $_1$ i $_2$ $_3$LB $_2$ ² iZjZf_lju ^b_jkbnbdZpbb dexqZ
SQXP $_3$B ² ihjy^dh\uc ghf_j khh[s_gby aZrbnjh\Zggh]h gZ ^Zgghf dexq_
b3180 y\eyxlky ij_^klZ\e_gb_f \ k_l_\hf ihjy^di_ [Zc SQXP kh gl_l
kl_ggh kf jbkmghd KmffZjgZy ^ebgZ _dlhjZ bgbpbZebaZpbb ² [Zcl

i_1	I_2	I_3
	PNUM	

Jbkmghd ² NhjfZl _dlhjZ bgbpbZebaZpbb

AgZq_gb_ _dlhjZ bgbpbZebaZpbb ^he`gh [ulv mgbdZevguf ^ey dZ`^h aZsbs_ggh]h kh_^bg_gby Lh _klv ^ey ^he`gh]h [ulv mgbdZevguf [ki_q_gh jZaebqb_ _ bgbpbZebaZpbb 9=ibG $_2$,G $_3$,G 3180 ^ey \k_o i_j_^Z\Z_fuo iZd_lh\
Dexq aZsblu khh[s_gby
Dexq rbnjh\Zgby b bfblhaZsblu khh[s_gby \uqbkey_lky ke_^mxsbf h[jZa

$$K_{PVJ}=ESPTREE\ .\ _1L_2L_3L$$

]^_ K ² dhjg_\hc dexq ljZgknhjfZ ENCR
i $_1$ B $_1$ i $_2$ $_3$L B $_2$ ² iZjZf_lju ^b_jkbnbdZpbb dexqZ i_j_^Z\Z_fu_ \ _dlhj_ b kf
H^ghjZah\uc _dlhj QRQFH
>ey aZsblu khh[s_gby bkihevam_lky h^ghjZah\uc _dlhj NhjfZl h^ghjZa hl bkihevam_fh]h ljZgknhjfZ
>ey ljZgknhjfZ ENCR_KUZNYECHIK_MGM_KTREE ENCR_KUZNYECHIK_MGM_MAC_KTREE EF^gh jZah\uc _dlhj QRQFH bf__l jZaf_j [bl b nhjfbjm_lky dZd ihdZaZgh gZ

zero	*PNUM*
salt	

Jbkmghd ² NhjfZl h^ghjZah\h]h _dlhjZ ENCR_KUZNYECHIK_MGM_KTREE
bENCR_KUZNYECHIK_MGM_MAC_KTREE

]^_]HUR ² [bl gme_c $_1$]HUR B
3180 ² ihjy^dh\uc ghf_j khh[s_gby [blZ ba _dlhjZ bgbpbZebaZpbb
VDOWhev [bl [_j_lky ba dexqZ ljZgknhjfZ K HQRU
>ey ljZgknhjfZ ENCR_MAGMA_MGM_KTREE ENCR_MAGMA_MGM_MAC_KTREE EF^ghjZah\uc _d lhj QRQFH bf__l jZaf_j [blZ b nhjfbjm_lky dZd ihdZaZgh gZ jbkmgd_

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J ²

zero	*PNUM*
salt	

Jbkmghd ² NhjfZl h^ghjZah\h]h _dlhjZ ENCR_KUZNYECHIK_MGM_KTREE ENCR_MAGMA_MGM_KTREE ENCR_MAGMA_MGM_ MAC_KTREE

]^_]HUR ² [bl gme_c]HUR B 1

3180 ² ihjy^dh\uc ghf_j khh[s_gby [blZ ba _dlhjZ bgbpbZebaZpbb kf j

VDOW ² khev [blZ [_j_lky ba dexgZ klfjZgknhjfZ K HQ=U

Bgufb keh\Zfb h^ghjZah\uc QR Q fdhlj key_lky gZ hkgh\Zgby VDOW %Wihdb, NOHQ 32

jy^dh\h]h ghf_jZ iZd_lZ 3180 ^Bmxsbf h[jZahf 3

QRQ=F]HUR G 3180 G VDOW

>ey ljZgknhjfh\ ENCR_KUZNYECHIK_MGM_KTREE ENCR_KUZNYECHIK_MGM_MAC_KTREE E^gh jZah\uc _dlhjZ QR Q fdhl__l jZaf_j [bl ^ey ljZgknhjfh\ ENCR_MAGMA_MGM_KTREE ENCR_ MAGMA_MGM_MAC_KTREE E^ghjZah\uc _dlhjZ QR Q fdhl jZaf_j [blZ NhjfZl $$'

NhjfZl $$' ^hihegbl_evguo bfblhaZsbsZ_fuo ^Zgguo aZ\bkbl hl ijbf_gy_fh] rbnjh\Zgby b hl lh]h bkihevam_lky eb (61 jZkrbj_gguc ghf_j iZd_lZ beb g_l gby ljZgknhjfh\ ENCR_KUZNYECHIK_MGM_KTREE ENCR_MAGMA_MGM_KTREE [_a bkihevah\Zgby (61 $$' nhjfbjm_lky dZd ihdZaZgh gZ jbkmgd_

Security Parameter Index (SPI)
Sequence Number (32 бита)

Jbkmghd ² NhjfZl $$' ^ey ljZgknhjfh\ ENCR_KUZNYECHIK_MGM_KTREE bENCR_MAGMA_MGM_KTREE [_a (61

>ey wlbo `_ ljZgknhjfh\ \ kemqZ_ bkihevah\Zgby (61 $$' nhjfbjm_lky dZd i d_

Security Parameter Index (SPI)
Extended Sequence Number (64 бита)

Jbkmghd ² NhjfZl $$' ^ey ljZgknhjfh\ ENCR_KUZNYECHIK_MGM_KTREE ENCR_MAGMA_MGM_KTREE [_a (61

< kemqZ_ ijbf_g_gby ljZgknhjfh\ ENCR_KUZNYECHIK_MGM_MAC_KTREE ENCR_MAGMA_MGM_ MAC_KTREE[_a bkihevah\Zgby (61 $$' nhjfbjm_lky dZd ihdZaZgh gZ jbkmgd_

Security Parameter Index (SPI)		
Sequence Number (32 бита)		
IV (64 бита)		
Payload Data		
Padding (0–255 байт)		
	Pad Length	Next Header

Jbkmghd ² NhjfZl $$' ^ey ljZgknhjfh\ ENCR_KUZNYECHIK_MGM_MAC_KTREE bENCR_MAGMA_MGM_MAC_KTREE [_a (61

>ey wlbo `_ ljZgknhjfh\ \ kemqZ_ bkihevah\Zgby (61 $$' nhjfbjm_lky dZd ihdZa

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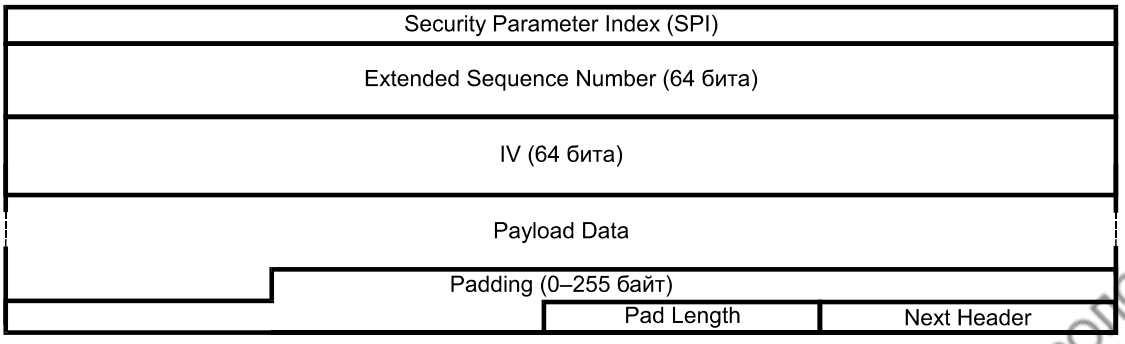
J ²



Jbkmghd ² NhjfZl $$' ^ey ljZgkrhkUZNYECHIK_MGM_MAC_KTREE bENCR_MAGMA_MGM_MAC_KTREE 61

LZ[ebpZ hibku\Z_l nhjfbjh\Zgb_ $$' \ aZ\bkbfhklb hl aZ^Zgguo iZjZf_ kh_^bg_gby

L Z [e b p Z ² NhjfZl $$'

LjZgknhjf ENCR	LjZgknhjf ESN	$$'
ENCR_KUZNYECHIK_MGM_KTREE ENCR_MAGMA_MGM_KTREE	noESN	63,G61
ENCR_KUZNYECHIK_MGM_KTREE ENCR_MAGMA_MGM_KTREE	ESN	63,G(61
ENCR_KUZNYECHIK_MGM_MAC_KTREE ENCR_MAGMA_MGM_MAC_KTREE	noESN	63,G61G,9G3D\OHG(63 'DWDOHU
ENCR_KUZNYECHIK_MGM_MAC_KTREE ENCR_MAGMA_MGM_MAC_KTREE	ESN	63,G(61G,9G3D\OHG(63 'DWDOHU

< kemqZ_ bkihevah\Zgby jZkrbj_gguo ghf_jh\ iZd_lh\ \ $$' \dexqZxlky iZd_lZ h^gZdk61ih\ekhklZ_ (63 iZd_lZ \dexqZ_l lhevdh (61eZZ^rkldZjlbl_ [blZ g_ i_j_^Zxlky

l j b f _ q Z g b _ ² AZf_lbf qlh ^ey ENCR_KUZNYECHIK_MGM_KTREE bENCR_MAGMA_MGM_ KTREE $$' g_ \dexqZ_l \ k_[y ,9 lZd dZd bfblhaZsblZ ,9 ijhbkoh^bl g_y\guf h[jZahf dhgdj_lguc dexq aZsblu khh[s_gby b h^ghjZah\uc _dlhj ijb bo g_khhl_lkl\bb bko [m^_l mki_rgh ijh_j_g gZ ijb_fghc klhjhg_ H^gZdk ENCR_KUZNYECHIK_MGM_MAC_KTREE bENCR_MAGMA_MGM_MAC_KTREE $$' \dexqZ_l \ k_[y ,9 qlh[u $$' ij_^klZ\eye kh[hc h^bg g_jZ ^Zgguo

<u[hj iZjZf_ljh\ ^b_jkbnbdZpbb dexqZ

<u[hj agZq_gbc iZjZf_ljh\ ^b_jkbnbdZpbb p_ebdhf hij_^_ey_lky klhjhgh khh[s_gb_ ljb_fgZy klhjhgZ bkihevam_l ihemq_ggu_ \ _dlhj_ bgbpbZebaZ ^ey \uqbke_gby dexqZ aZsblu khh[s_gby DZ`^Zy klhjhgZ \u[bjZ_l iZjZf_lju hlkueZ_fuo _x khh[s_gbc g_aZ\bkbfh hl ^jm]hc klhjhgu

GZ]jmadZ gZ djbilh]jZnbq_kdb_ dexqb

lj_^iheZ]Z_lky qlh i_j_^ZxsZy klhjhgZ _^_l mq_l h[t_fZ ^Zgguo aZ dexqK_PVJ dexq hij_^_ey_lky agZq_gby ?kb ijb h[jZ[hld_ hq_j_^gh]h ,3 iZd f ljZnbdZ [m^_l ij_\ur_g eb[h ijhbahc^_l i_j_ihe SeQXgb_lhk^_l mq_l [jZZ[hldb ^Zg iZd_lZ \ujZ[Zlu\Z_lky ghKuc imelx f m_ebq_gby iZjZf_lj_Z^bgbpm SQXiPb wlhf h[gmey_lky ljb \hafh`ghf ij_\ur_gbb gZ]jmadb gZ dexq \lhjh]h mjh\gy be iZjZf_lij m\_ebqb\Z_lky gZ _^bgbpm Z ijb \hafh`ghf ij_\ur_gbb gZ]jmadb g beb ijb i_j_ihe gZjZf_lj m_ebqb\Z_lky gZ _^bgbpm ljb kih_l m_fZbdkbfZev gh]h agZq_gby ^Zggh_ aZsbs_ggh_ kh_^bg_gb_ m^Zey_lky b \aZf_g g_]l dexqhf_K_HQFU

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J ²

SQXP $_3 = i_3$ L

_keb i_3 PD[lh $_3$ L $_2 = i_2$

_keb i_2 PD[lh $_2$ L $_1 = i_1$

keb i_1 PD[lh g bkihevah\Zlv 6$

bgZq $_{PVJ}$ K= ESPTREE . $_1$ L $_2$ L $_3$ L

]^_ agZq i_1 gby i_2 PD[i_3 PD[hij_^_eyxlky iheblbdhc [_ahiZkghklb gh \ ex[hf kemqZ_ l
_ebqbgZ $_1$ fb PD[" $_2$ PD[" $_3$ ib PD["

l_j_^ZxsZy klhjhgZ g_ ^he`gZ i_jK kqblZgZ\h ijb hlijZ\d_ hq_j_^gh]h iZd_l
ihke_^g__ agZq_gb_ dexqZ ^he`gh khojZgylvky b bkihevah\Zlvky _keb agZq_g
nbdZpbb $_1$ $_2$ $_3$ g_ baf_gbebkv ih kjZ\g_gbx k ij_^u^msbf hlijZ\e_gguf iZd_lhf Wlh
b ^ey ijhf_`mlhqguo dexq_c \ ^_j__

lj_^iheZ]Z_lky qlh gZ ijb_fghc klhjhg_ lZd`_ _^_lky mq_lh[t_fZ ^Zgguo a
ghf dexq $_{PVJ}$ dexq hij_^_ey_lky agZq $_1$ $_2$ $_3$ gby ?keb ijb ijboh^_ hq_j_^gh]h (63 iZd_lZ h
_f ljZnbdZ aZsbs_ggh]h gZ ^Zgghf ^ dexq $_{PVJ}$ dexq g_l fZdkbfZevgh ^himklbfh]h ^ey Z[h
agZq_gby gZ]jmadb gZ dexq beb ijhbahc^_l ij_\ur_gb_ gZ]jmadb gZ ijhf_`mlhq
h[jZ[hldZ iZd_lZ ijh^he`Z_lky dZd h[uqgh gh ijb wlhf bgbpbbjm_lky kha^Zgb_
kh_^bg_gby k gh\uf dexq $_{HQF-U}$ dexq hf lhjh_ ^he`gh bkihevah\Zlvky \aZf_g l_dms_]h

Ijb_fgZy klhjhgZ g_ ^he`gZ i_jK kqblZgZ\h ijb ihemq_gbb dZ`^h]h gh\h]h iZ
ihke_^g__ agZq_gb_ dexqZ ^he`gh khojZgylvky b bkihevah\Zlvky _keb agZq_
kbnbdZpbb g_ baf_gbebkv ih kjZ\g_gbx k ij_^u^msbfb iZd_lZfb J_dhf_g^m_lky
ihke_^gbo agZq_gbc qlh[u g_ i_j_kqblu\Zlv dexqb ebrgbc jZa ijb gZjmr_gbb i
iZd_lh\ \ k_lb Wlh kijZ_^eb\h b ^ey ijhf_`mlhqguo dexq_c \ ^_j__

?keb hlkueZ_fuc ,3 iZd_l gZklhevdh _ebd qlh _]h jZaf_j ij_\hkoh^bl fZdk
fmx gZ]jmadm gZ dexq lh \ lmgg_evghf j_`bf_ i_j_^ZxsZy klhjhgZ ^he`gZ njZ]
g_kdhevdh ,3 njZ]f_glh\ lZdbf h[jZahf qlh[u jZaf_j ex[h]h njZ]f_glZ g_ ij_\hkol
gZ]jmadm gZ dexq b ^Ze__ ^_ckl\h\Zlv ih Ze]hjblfm mq_lZ gZdhie_ggh]h h[t_f
imgdl_ < ljZgkihjlghf j_`bf_ i_j_^ZqZ lZdh]h ,3 iZd_lZ y\ey_lky g_\hafh`ghc

?keb gZ ijb_fghc klhjhg_ nbdkbjm_lky ihemq_gb_ agZqbl_evgh]h qbkeZ bk
dhjhldbc ijhf_`mlhd \j_f_gb \ dhgl_dkl_ dZdh]h lh aZsbs_ggh]h kh_^bg_gby l
ijbagZdhf ijh_^_gby ZlZdb hkh[_ggh _keb ijb wlhf gZ[ex^Z_lky oZ $_1$ lbqgh_ ba
i_2 i_3 bSQXP ihemqZ_fuo iZd_lZo < wlhf kemqZ_ ijb_fgZy klhjhgZ fh`_l m^Zeblv ^
gh_ kh_^bg_gb_ b kha^Zlv b^_glbqgh_ _fm gh k gh\uf agZq $_{HQF-U}$ gb_f 63, b k gh\u

l j b f _ q Z g b _ ² < lhf kemqZ_ _keb lZdh_ ih_^_gb_ ^_ckl\bl_evgh \ua\Zgh Zl
aZsbs_ggh]h kh_^bg_gby k gh\uf agZq_gb_f 63, fh`_l kihkh[kl\h\Zlv ijhlb\h^_ckl\bx wlh
jmrbl_ev ^he`_g agZlv 63, ^ey __ ijh\_^_gby < aZ\bkbfhklb hl \hafh`ghkl_c gZjmrbl_ey
[ulv ljb\bZevghc aZ^Zq_c

B^_glbnbdZlhju Ze]hjblfh\

Gb`_ ijb_^_gu b^_glbnbdZlhju jhkkbckdbo Ze]hjblfh\ bkihevam_fuo \ gZkl
dZpbyo Wlb b^_glbnbdZlhju bkihevamxlky \ ijhlhdhe_ ,.(Y ijb kh]eZkh\Zgbb i
B^_glbnbdZlhju ^ey ljZgknhjf ENCR ijb ^_^_gu \ lZ[ebp_

L Z [e b p Z ² B^_glbnbdZlhju jhkkbckdbo Ze]hjblfh\

LjZgknhjf ENCR	B^_glbnbdZlhj
ENCR_KUZNYECHIK_MGM_KTREE	32
ENCR_MAGMA_MGM_KTREE	33
ENCR_KUZNYECHIK_MGM_MAC_KTREE	34
ENCR_MAGMA_MGM_MAC_KTREE	35

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J ²

ljbeh`_gb_ $
kijZ\hqgh_

FZdkbfZevguc h[t_f h[jZ[Zlu\Z_fh]h ljZnbdZ

LZ[ebpZ : kh^_j`bl j_dhf_g^m_fu_ agZq_gby fZdkbfZevgh ^himklbfh]h kmffZ
dhlhjuc h[jZ[Zlu\Z_lky gZ h^ghf_dhx^hklb`_gbb fZdkbfZevgh]h h[t_fZ ^he`_g \ujZ[
gh\uc dexq_kba ^_j_\Z dexq_c ih ijhp_^mj_ hibkZgghc \

LZ[ebpZ : ² FZdkbfZevgh ^himklbfuc h[t_f ljZnbdZ h[jZ[Zlu\Z_fh]h gZ h^ghf

LjZgknhjfENCR	FZdkbfZevguc h[t_f ljZnbdZ
ENCR_KUZNYECHIK_MGM_KTREE ENCR_KUZNYECHIK_MGM_MAC_KTREE	2[41] [Zcl
ENCR_MAGMA_MGM_KTREE ENCR_MAGMA_MGM_MAC_KTREE	2 [Zcl

ljbf_qZgby
MdZaZggu_ agZq_gby ghkyl j_dhf_g^Zl_evguc oZjZdl_j
Mqblu\Zy lh qlh fZdkbfZevgh_ qbkeh iZd_lh\ aZsbsZ_fug_gZh h^ghf_dhx^hklb`_gbb
2[24] Z fZdkbfZevguc jZaf_j ,3 iZd_lZ g_[iZ_lurZlh h]jZgbq_gb_ ^_ckl\m_l dZd gZ a
iZd_l lZd b gZ ,3 iZd_l ihemqZxsbcky \ j_amevlZl_ ijbf_g_gby d aZsbsZ_fhfm iZd
guc h[t_f ljZnbdZ aZsbsZ_fh]h gZ h^ghf_dhx^hklb`_gbb
gb_[41] [Zcl ^ey ljZgknhjfENCR_KUZNYECHIK_MGM_KTREE ENCR_KUZNYECHIK_MGM_MAC_KTREE ey_lky
g_^hklb`bfuf

ljb bkihevah\Zgbb ljZgknhjfh\ [Zabjmxsboky gZ rbnj_ ©FZ]fZª ENCR_MAGMA_MG
MAGMA_MGM_MAC_KTREEj_dhf_g^m_lky h]jZgbqb\Zlv dhebq_kl\h Khjm^ehfuc dexq_c
\h]h dexq_c agZq_gb_f f_gvrbf q_f jZaf_j ^_j_\Z dexq_c ijbi ihehgh gZ ijbf_hj_ iml_f
_^_gby h]jZgbq_gbc gZ fZdkbfZevgu_ agZq_gby wlbo kq_lqbdh\

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Ijbeh`_gb_ ;
kijZ\hqgh_

L_klh\u_ _dlhju

>Zggh_ ijbeh`_gb_ kh^_j`bl l_klh\u_ _dlhju AgZq_gby l_klh\uo _dlhjh\ ijb_^_gu jbqghf \b^_

; LjZgknhjf **ENCR_KUZNYECHIK_MGM_KTREE, пример 1**

K:
b6 18 0c 14 5c 51 2d bd 69 d9 ce a9 2c ac 1b 5c
e1 bc fa 73 79 2d 61 af 0b 44 0d 84 b5 22 cc 38
i1 = 00, i2 = 0000, i3 = 0000, pnum = 000000
K_{msg}:
2f f1 c9 0e de 78 6e 06 1e 17 b3 74 d7 82 af 7b
d8 80 bd 52 7c 66 a2 ba dc 3e 56 9a ab 27 1d a4
salt [12]:
7b 67 e6 f2 44 f9 7f 06 78 95 2e 45
nonce [16]:
00 00 00 00 7b 67 e6 f2 44 f9 7f 06 78 95 2e 45
IV [8]:
00 00 00 00 00 00 00 00
AAD [8]:
51 46 53 6b 00 00 00 01
plaintext [64]:
45 00 00 3c 23 35 00 00 7f 01 ee cc 0a 6f 0a c5
0a 6f 0a 1d 08 00 f3 5b 02 00 58 00 61 62 63 64
65 66 67 68 69 6a 6b 6c 6d 6e 6f 70 71 72 73 74
75 76 77 61 62 63 64 65 66 67 68 69 01 02 02 04
ciphertext [64]:
18 9d 12 88 b7 18 f9 ea be 55 4b 23 9b ee 65 96
c6 d4 ea fd 31 64 96 ef 90 1c ac 31 60 05 aa 07
62 97 b2 24 bf 6d 2b e3 5f d6 f6 7e 7b 9d eb 31
85 ff e9 17 9c a9 bf 0b db af c2 3e ae 4d a5 6f
ESP ICV [12]:
50 b0 70 a1 5a 2b d9 73 86 89 f8 ed
ESP packet [112]:
45 00 00 70 00 4d 00 00 ff 32 91 4f 0a 6f 0a c5
0a 6f 0a 1d 51 46 53 6b 00 00 00 01 00 00 00 00
00 00 00 00 18 9d 12 88 b7 18 f9 ea be 55 4b 23
9b ee 65 96 c6 d4 ea fd 31 64 96 ef 90 1c ac 31
60 05 aa 07 62 97 b2 24 bf 6d 2b e3 5f d6 f6 7e
7b 9d eb 31 85 ff e9 17 9c a9 bf 0b db af c2 3e
ae 4d a5 6f 50 b0 70 a1 5a 2b d9 73 86 89 f8 ed

; LjZgknhjf **ENCR_KUZNYECHIK_MGM_KTREE**ijbf_j

K:
b6 18 0c 14 5c 51 2d bd 69 d9 ce a9 2c ac 1b 5c
e1 bc fa 73 79 2d 61 af 0b 44 0d 84 b5 22 cc 38
i1 = 00, i2 = 0001, i3 = 0001, pnum = 000000
K_{msg}:
9a ba c6 57 78 18 0e 6f 2a f6 1f b8 d5 71 62 36
66 c2 f5 13 0d 54 e2 11 6c 7d 53 0e 6e 7d 48 bc
salt [12]:
7b 67 e6 f2 44 f9 7f 06 78 95 2e 45
nonce [16]:
00 00 00 00 7b 67 e6 f2 44 f9 7f 06 78 95 2e 45
IV [8]:
00 00 01 00 01 00 00 00
AAD [8]:
51 46 53 6b 00 00 00 10

ФЕДЕРАЛЬНОЕ АГЕНТСТВО ПО ТЕХНИЧЕСКОМУ РЕГУЛИРОВАНИЮ И МЕТРОЛОГИИ

J 2

plaintext [64]:
45 00 00 3c 23 48 00 00 7f 01 ee b9 0a 6f 0a c5
0a 6f 0a 1d 08 00 e4 5b 02 00 67 00 61 62 63 64
65 66 67 68 69 6a 6b 6c 6d 6e 6f 70 71 72 73 74
75 76 77 61 62 63 64 65 66 67 68 69 01 02 02 04
ciphertext [64]:
78 0a 2c 62 62 32 15 7b fe 01 76 32 f3 2d b4 d0
a4 fa 61 2f 66 c2 bf 79 d5 e2 14 9b ac 1d fc 4b
15 4b 69 03 4d c2 1d ef 20 90 6d 59 62 81 12 7c
ff 72 56 ab f0 0b a1 22 bb 5e 6c 71 a4 d4 9a 4d
ESP ICV [12]:
c2 2f 87 40 83 8e 3d fa ce 91 cc b8
ESP packet [112]:
45 00 00 70 00 5c 00 00 ff 32 91 40 0a 6f 0a c5
0a 6f 0a 1d 51 46 53 6b 00 00 00 10 00 00 01 00
01 00 00 00 78 0a 2c 62 62 32 15 7b fe 01 76 32
f3 2d b4 d0 a4 fa 61 2f 66 c2 bf 79 d5 e2 14 9b
ac 1d fc 4b 15 4b 69 03 4d c2 1d ef 20 90 6d 59
62 81 12 7c ff 72 56 ab f0 0b a1 22 bb 5e 6c 71
a4 d4 9a 4d c2 2f 87 40 83 8e 3d fa ce 91 cc b8

; L j Z g k n h j f **ENCR_MAGMA_MGM_KTREė** j b f _ j

K:
5b 50 bf 33 78 87 02 38 f3 ca 74 0f d1 24 ba 6c
22 83 ef 58 9b e6 f4 6a 89 4a a3 5d 5f 06 b2 03
i1 = 00, i2 = 0000, i3 = 0000, pnum = 000000
K_{msg}:
25 65 21 e2 70 b7 4a 16 4d fc 26 e6 bf 0c ca 76
5e 9d 41 02 7d 4b 7b 19 76 2b 1c c9 01 dc de 7f
salt [4]:
cf 36 63 12
nonce [8]:
00 00 00 00 cf 36 63 12
IV [8]:
00 00 00 00 00 00 00 00
AAD [8]:
c8 c2 b2 8d 00 00 00 01
plaintext [64]:
45 00 00 3c 24 2d 00 00 7f 01 ed d4 0a 6f 0a c5
0a 6f 0a 1d 08 00 de 5b 02 00 6d 00 61 62 63 64
65 66 67 68 69 6a 6b 6c 6d 6e 6f 70 71 72 73 74
75 76 77 61 62 63 64 65 66 67 68 69 01 02 02 04
ciphertext [64]:
fa 08 40 33 2c 4f 3f c9 64 4d 8c 2c 4a 91 7e 0c
d8 6f 8e 61 04 03 87 64 6b b9 df bd 91 50 3f 4a
f5 d2 42 69 49 d3 5a 22 9e 1e 0e fc 99 ac ee 9e
32 43 e2 3b a4 d1 1e 84 5c 91 a7 19 15 52 cc e8
ESP ICV [8]:
5f 4a fa 8b 02 94 0f 5c
ESP packet [108]:
45 00 00 6c 00 62 00 00 ff 32 91 3e 0a 6f 0a c5
0a 6f 0a 1d c8 c2 b2 8d 00 00 00 01 00 00 00 00
00 00 00 00 fa 08 40 33 2c 4f 3f c9 64 4d 8c 2c
4a 91 7e 0c d8 6f 8e 61 04 03 87 64 6b b9 df bd
91 50 3f 4a f5 d2 42 69 49 d3 5a 22 9e 1e 0e fc
99 ac ee 9e 32 43 e2 3b a4 d1 1e 84 5c 91 a7 19
15 52 cc e8 5f 4a fa 8b 02 94 0f 5c

; L j Z g k n h j f **ENCR_MAGMA_MGM_KTREė** j b f _ j

K:
5b 50 bf 33 78 87 02 38 f3 ca 74 0f d1 24 ba 6c
22 83 ef 58 9b e6 f4 6a 89 4a a3 5d 5f 06 b2 03
i1 = 00, i2 = 0001, i3 = 0001, pnum = 000000

ФЕДЕРАЛЬНОЕ АГЕНТСТВО ПО ТЕХНИЧЕСКОМУ РЕГУЛИРОВАНИЮ И МЕТРОЛОГИИ

K_{msg}:
20 e0 46 d4 09 83 9b 23 f0 66 a5 0a 7a 06 5b 4a
39 24 4f 0e 29 ef 1e 6f 2e 5d 2e 13 55 f5 da 08
salt [4]:
cf 36 63 12
nonce [8]:
00 00 00 00 cf 36 63 12
IV [8]:
00 00 01 00 01 00 00 00
AAD [8]:
c8 c2 b2 8d 00 00 00 10
plaintext [64]:
45 00 00 3c 24 40 00 00 7f 01 ed c1 0a 6f 0a c5
0a 6f 0a 1d 08 00 cf 5b 02 00 7c 00 61 62 63 64
65 66 67 68 69 6a 6b 6c 6d 6e 6f 70 71 72 73 74
75 76 77 61 62 63 64 65 66 67 68 69 01 02 02 04
ciphertext [64]:
7a 71 48 41 a5 34 b7 58 93 6a 8e ab 26 91 40 a8
25 a7 f3 5d b9 e4 37 1f e7 6c 99 9c 9b 88 db 72
1d c7 59 f6 56 b5 b3 ea b6 b1 4d 6b d7 7a 07 1d
4b 93 78 bd 08 97 6c 33 ed 9a 01 91 bf fe a1 dd
ESP ICV [8]:
dd 5d 50 9a fd b8 09 98
ESP packet [108]:
45 00 00 6c 00 71 00 00 ff 32 91 2f 0a 6f 0a c5
0a 6f 0a 1d c8 c2 b2 8d 00 00 00 10 00 00 01 00
01 00 00 00 7a 71 48 41 a5 34 b7 58 93 6a 8e ab
26 91 40 a8 25 a7 f3 5d b9 e4 37 1f e7 6c 99 9c
9b 88 db 72 1d c7 59 f6 56 b5 b3 ea b6 b1 4d 6b
d7 7a 07 1d 4b 93 78 bd 08 97 6c 33 ed 9a 01 91
bf fe a1 dd dd 5d 50 9a fd b8 09 98

; L j Z g k n h j f **ENCR_KUZNYECHIK_MGM_MAC_KTREE** i j b f _ j

K:
98 bd 34 ce 3b e1 9a 34 65 e4 87 c0 06 48 83 f4
88 cc 23 92 63 dc 32 04 91 9b 64 3f e7 57 b2 be
i1 = 00, i2 = 0000, i3 = 0000, pnum = 000000
K_{msg}:
98 f1 03 01 81 0a 04 1c da dd e1 bd 85 a0 8f 21
8b ac b5 7e 00 35 e2 22 c8 31 e3 e4 f0 a2 0c 8f
salt [12]:
6c 51 cb ac 93 c4 5b ea 99 62 79 1d
nonce [16]:
00 00 00 00 6c 51 cb ac 93 c4 5b ea 99 62 79 1d
IV [8]:
00 00 00 00 00 00 00 00
AAD [80]:
3d ac 92 6a 00 00 00 01 00 00 00 00 00 00 00 00
45 00 00 3c 0c f1 00 00 7f 01 05 11 0a 6f 0a c5
0a 6f 0a 1d 08 00 48 5c 02 00 03 00 61 62 63 64
65 66 67 68 69 6a 6b 6c 6d 6e 6f 70 71 72 73 74
75 76 77 61 62 63 64 65 66 67 68 69 01 02 02 04
plaintext [0]:
ciphertext [0]:
ESP ICV [12]:
ca c5 8c e5 e8 8b 4b f3 2d 6c f0 4d
ESP packet [112]:
45 00 00 70 00 01 00 00 ff 32 91 9b 0a 6f 0a c5
0a 6f 0a 1d 3d ac 92 6a 00 00 00 01 00 00 00 00
00 00 00 00 45 00 00 3c 0c f1 00 00 7f 01 05 11
0a 6f 0a c5 0a 6f 0a 1d 08 00 48 5c 02 00 03 00
61 62 63 64 65 66 67 68 69 6a 6b 6c 6d 6e 6f 70
71 72 73 74 75 76 77 61 62 63 64 65 66 67 68 69
01 02 02 04 ca c5 8c e5 e8 8b 4b f3 2d 6c f0 4d

ФЕДЕРАЛЬНОЕ АГЕНТСТВО ПО ТЕХНИЧЕСКОМУ РЕГУЛИРОВАНИЮ И МЕТРОЛОГИИ

; L j Z g k n h j f **ENCR_KUZNYECHIK_MGM_MAC_KTREE** j b f _ j

K:
98 bd 34 ce 3b e1 9a 34 65 e4 87 c0 06 48 83 f4
88 cc 23 92 63 dc 32 04 91 9b 64 3f e7 57 b2 be
i1 = 00, i2 = 0000, i3 = 0001, pnum = 000000
K_{msg}:
02 c5 41 87 7c c6 23 f3 f1 35 91 9a 75 13 b6 f8
a8 a1 8c b2 63 99 86 2f 50 81 4f 52 91 01 67 84
salt [12]:
6c 51 cb ac 93 c4 5b ea 99 62 79 1d
nonce [16]:
00 00 00 00 6c 51 cb ac 93 c4 5b ea 99 62 79 1d
IV [8]:
00 00 00 00 01 00 00 00
AAD [80]:
3d ac 92 6a 00 00 00 06 00 00 00 00 01 00 00 00
45 00 00 3c 0c fb 00 00 7f 01 05 07 0a 6f 0a c5
0a 6f 0a 1d 08 00 43 5c 02 00 08 00 61 62 63 64
65 66 67 68 69 6a 6b 6c 6d 6e 6f 70 71 72 73 74
75 76 77 61 62 63 64 65 66 67 68 69 01 02 02 04
plaintext [0]:
ciphertext [0]:
ESP ICV [12]:
ba bc 67 ec 72 a8 c3 1a 89 b4 0e 91
ESP packet [112]:
45 00 00 70 00 06 00 00 ff 32 91 96 0a 6f 0a c5
0a 6f 0a 1d 3d ac 92 6a 00 00 00 06 00 00 00 00
01 00 00 00 45 00 00 3c 0c fb 00 00 7f 01 05 07
0a 6f 0a c5 0a 6f 0a 1d 08 00 43 5c 02 00 08 00
61 62 63 64 65 66 67 68 69 6a 6b 6c 6d 6e 6f 70
71 72 73 74 75 76 77 61 62 63 64 65 66 67 68 69
01 02 02 04 ba bc 67 ec 72 a8 c3 1a 89 b4 0e 91

; L j Z g k n h j f **ENCR_MAGMA_MGM_MAC_KTREE** j b f _ j

K:
d0 65 b5 30 fa 20 b8 24 c7 57 0c 1d 86 2a e3 39
2c 1c 07 6d fa da 69 75 74 4a 07 a8 85 7d bd 30
i1 = 00, i2 = 0000, i3 = 0000, pnum = 000000
K_{msg}:
4c 61 45 99 a0 a0 67 f1 94 87 24 0a e1 00 e1 b7
ea f2 3e da f8 7e 38 73 50 86 1c 68 3b a4 04 46
salt [4]:
88 79 8f 29
nonce [8]:
00 00 00 00 88 79 8f 29
IV [8]:
00 00 00 00 00 00 00 00
AAD [80]:
3e 40 69 9c 00 00 00 01 00 00 00 00 00 00 00 00
45 00 00 3c 0e 08 00 00 7f 01 03 fa 0a 6f 0a c5
0a 6f 0a 1d 08 00 36 5c 02 00 15 00 61 62 63 64
65 66 67 68 69 6a 6b 6c 6d 6e 6f 70 71 72 73 74
75 76 77 61 62 63 64 65 66 67 68 69 01 02 02 04
plaintext [0]:
ciphertext [0]:
ESP ICV [8]:
4d d4 25 8a 25 35 95 df
ESP packet [108]:
45 00 00 6c 00 13 00 00 ff 32 91 8d 0a 6f 0a c5
0a 6f 0a 1d 3e 40 69 9c 00 00 00 01 00 00 00 00
00 00 00 00 45 00 00 3c 0e 08 00 00 7f 01 03 fa

ФЕДЕРАЛЬНОЕ АГЕНТСТВО ПО ТЕХНИЧЕСКОМУ РЕГУЛИРОВАНИЮ И МЕТРОЛОГИИ

0a 6f 0a c5 0a 6f 0a 1d 08 00 36 5c 02 00 15 00
61 62 63 64 65 66 67 68 69 6a 6b 6c 6d 6e 6f 70
71 72 73 74 75 76 77 61 62 63 64 65 66 67 68 69
01 02 02 04 4d d4 25 8a 25 35 95 df

; L j Z g k n h j f **ENCR_MAGMA_MGM_MAC_KTREE** j b f _ j

K:
d0 65 b5 30 fa 20 b8 24 c7 57 0c 1d 86 2a e3 39
2c 1c 07 6d fa da 69 75 74 4a 07 a8 85 7d bd 30
i1 = 00, i2 = 0000, i3 = 0001, pnum = 000000
K_{msg}:
b4 f3 f9 0d c4 87 fa b8 c4 af d0 eb 45 49 f2 f0
e4 36 32 b6 79 19 37 2e 1e 96 09 ea f0 b8 e2 28
salt [4]:
88 79 8f 29
nonce [8]:
00 00 00 00 88 79 8f 29
IV [8]:
00 00 00 00 01 00 00 00
AAD [80]:
3e 40 69 9c 00 00 00 06 00 00 00 00 01 00 00 00
45 00 00 3c 0e 13 00 00 7f 01 03 ef 0a 6f 0a c5
0a 6f 0a 1d 08 00 31 5c 02 00 1a 00 61 62 63 64
65 66 67 68 69 6a 6b 6c 6d 6e 6f 70 71 72 73 74
75 76 77 61 62 63 64 65 66 67 68 69 01 02 02 04
plaintext [0]:
ciphertext [0]:
ESP ICV [8]:
84 84 a9 23 30 a0 b1 96
ESP packet [108]:
45 00 00 6c 00 18 00 00 ff 32 91 88 0a 6f 0a c5
0a 6f 0a 1d 3e 40 69 9c 00 00 00 06 00 00 00 00
01 00 00 00 45 00 00 3c 0e 13 00 00 7f 01 03 ef
0a 6f 0a c5 0a 6f 0a 1d 08 00 31 5c 02 00 1a 00
61 62 63 64 65 66 67 68 69 6a 6b 6c 6d 6e 6f 70
71 72 73 74 75 76 77 61 62 63 64 65 66 67 68 69
01 02 02 04 84 84 a9 23 30 a0 b1 96

ФЕДЕРАЛЬНОЕ АГЕНТСТВО ПО ТЕХНИЧЕСКОМУ РЕГУЛИРОВАНИЮ И МЕТРОЛОГИИ

ljbeh`_gb_ <
kijZ\hqgh_

:e]hjblf h[jZ[hldb jZkrbj_ggh]h ghf_jZ iZd_lZ

:e]hjblf h[jZ[hldb jZkrbj_ggh]h ghf_jZ iZd_lZ gZ ijb_fghc klhjhg_ <k_ bkihev ij_^klZ\eyxl kh[hc [_aagZdh\u_ p_eu_ [blgu_ qbkeZ \k_ \uqbke_gbbykh_evZxhky ih fu_ i_j_f_ggu_ hibkZgu \

```
If (Rl >= Wsz — 1)
  If (Sl >= Rl - Wsz + 1)
    Sh = Rh
    If (Sl <= Rl)
                If (соответствующий Sl бит сброшен)
                    If (имитовставка корректная)
                        Установить соответствующий Sl бит
                        Продолжить обработку пакета
                    Else Отбросить пакет
                Else Отбросить пакет
    Else
                If (имитовставка корректная)
                    Rl = Sl (сдвинуть окно)
                    Установить соответствующий Sl бит
                    Продолжить обработку пакета
                Else Отбросить пакет
  Else
    Sh = Rh + 1
            If (имитовставка корректная)
                Rl = Sl (сдвинуть окно)
      Rh = Rh + 1
                Установить соответствующий Sl бит
                Продолжить обработку пакета
            Else Отбросить пакет
Else
  If (Sl >= Rl - Wsz + 1)
    Sh = Rh - 1
            If (соответствующий Sl бит сброшен)
                If (имитовставка корректная)
                    Установить соответствующий Sl бит
                    Продолжить обработку пакета
                Else Отбросить пакет
            Else Отбросить пакет
  Else
    Sh = Rh
    If (Sl <= Rl)
                If (соответствующий Sl бит сброшен)
                    If (имитовставка корректная)
                        Установить соответствующий Sl бит
                        Продолжить обработку пакета
                    Else Отбросить пакет
                Else Отбросить пакет
    Else
                If (имитовставка корректная)
                    Rl = Sl (сдвинуть окно)
                    Установить соответствующий Sl бит
                    Продолжить обработку пакета
                Else Отбросить пакет
```

ФЕДЕРАЛЬНОЕ АГЕНТСТВО ПО ТЕХНИЧЕСКОМУ РЕГУЛИРОВАНИЮ И МЕТРОЛОГИИ

J 2

;b[ebh]jZnby

> @ ,(7) 5)& K D_gl AZsbs_ggh_ \eh`_gb_ ,3 (63 .HQW 6 ©,3 (QFDSVXODWLQJ
5)& '2, 5)& 'HFHPEHU KWWS ZZZ UIF HGLWRU R

ФЕДЕРАЛЬНОЕ АГЕНТСТВО ПО ТЕХНИЧЕСКОМУ РЕГУЛИРОВАНИЮ И МЕТРОЛОГИИ

M>D HDK

Dexq_\u_ keh\Z djbilh]jZnbq_kdb_ ijhlhdheu j_`bfu rbnjh\Zgb_ bfblh\kl

ФЕДЕРАЛЬНОЕ АГЕНТСТВО ПО ТЕХНИЧЕСКОМУ РЕГУЛИРОВАНИЮ И МЕТРОЛОГИИ

ФЕДЕРАЛЬНОЕ АГЕНТСТВО ПО ТЕХНИЧЕСКОМУ РЕГУЛИРОВАНИЮ И МЕТРОЛОГИИ

J_^Zdlhj E < Dhj_lgbdh\Z
L_ogbq_kdbc j_^Zdlhj < G ljmkZdh\Z
Dhjj_dlhj E K Euk_gdh
Dhfivxl_jgZy _jkldZ ? H :klZrbgZ

K^Zgh \ gZ[hj lh^ibkZgh \ i_qZlv [1] NZjgZlmjZ :jbZe
Mke i_q e Mq ba^ e
lh^]hlh\e_gh gZ hkgh_ we_dljhgghc _jkbb ij_^hklZ\e_gghc jZajZ[hlqbdhf klZg^Zjl

---

Kha^Zgh \ _^bgbqghf bkiheg_gbb \h N=MI ©KL:G>:JLBGNHJFª ^ey dhfie_dlh\Zgby N_^_jZevgh
nhg^Z klZg^Zjlh\ Fhkd\Z GZobfh\kdbc ij l ^ d
ZZZ JRVWLQIR UX LQIR#JRVWLQIR UX

ФЕДЕРАЛЬНОЕ АГЕНТСТВО ПО ТЕХНИЧЕСКОМУ РЕГУЛИРОВАНИЮ И МЕТРОЛОГИИ